

HAMMAR IPF — SMART SCADA

Advanced GOSP Simulation & Engineering Platform

COMPLETE CODEBASE DOCUMENTATION

Developed and Modeled by Petroleum Engineering Students
Al-Maaqal University

Under the Direct Supervision of:
Dr. Eng. Saher Adel & Dr. Mahmoud Badawy

Generated automatically on 2026-05-20

■ File: README.md

```
# Hammar IPF - Smart SCADA & Engineering Dashboard ■■ ![Hammar
SCADA](https://img.shields.io/badge/Platform-Smart%20SCADA-blue) ![API
Standards](https://img.shields.io/badge/Standards-API%2012J%20%7C%20ASTM-green) An advanced GOSP (Gas Oil
Separation Plant) simulation and engineering platform designed for real-time monitoring and calculation of
oil and gas surface facilities. --- ## ■ Key Features - **Interactive SVG Process Flow:** Real-time animated
schematic of the plant (Wellhead to Export). - **Engineering Calculators:** - 3-Phase Separator Sizing (API
12J / Souders-Brown). - Oil Treatment (Dehydration & Desalting). - Produced Water Treatment. - Gas Treatment
& Sweetening. - Chemical Injection Rates. - **Dynamic Simulation:** Real-time mass balance and performance
monitoring. - **PWA Ready:** Installable on mobile and desktop for offline monitoring. - **Bilingual
Support:** Full English & Arabic interface. - **Digital Twin:** Integrated 3D visualization of the GOSP
station. ## ■ Standards & Compliance This project adheres to international engineering standards: - **API
12J:** Specification for Oil and Gas Separators. - **ASTM D4007:** Standard Test Method for Water and
Sediment. - **GPSA:** Engineering Data Book for Gas Processing. - **API 14.1 / 14.5:** Gas Sampling &
Measurement. ## ■ How to Run 1. **Local Demo:** Run `start_online.vbs` to start the local server and create
a secure tunnel for global access. 2. **Web Version:** Open `index.html` in any modern browser. 3. **Data
Analysis:** Use the Streamlit version (`app.py`) for deep data analytics. --- ### ■ Developer **Sajad
Khano** [GitHub Profile](https://github.com/sajadkhano) --- *Disclaimer: This is an engineering simulation
tool. Always verify results with certified engineering software before field application.*
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■ File: index.html

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport"
content="width=device-width, initial-scale=1.0, maximum-scale=5.0, user-scalable=yes"> <meta
name="mobile-web-app-capable" content="yes"> <meta name="apple-mobile-web-app-capable" content="yes"> <meta
name="apple-mobile-web-app-status-bar-style" content="black-translucent"> <meta
name="apple-mobile-web-app-title" content="Hammar SCADA"> <meta name="theme-color" content="#1a2030"> <meta
name="msapplication-TileColor" content="#1a2030"> <title>Hammar IPF - Smart SCADA</title> <meta
name="description" content="Advanced GOSP simulation platform for oil separation, gas treatment, and surface
facilities engineering"> <link rel="manifest" href="manifest.json"> <link rel="apple-touch-icon"
href="icon-192.png"> <link rel="icon" type="image/png" href="icon-192.png"> <link href="https://fonts.google
apis.com/css2?family=Inter:wght@300;400;500;600;700;800;900&family=JetBrains+Mono:wght@400;500;600;700&displ
ay=swap" rel="stylesheet"> <script
src="https://cdn.jsdelivr.net/npm/chart.js@4.4.7/dist/chart.umd.min.js"></script> <script
src="https://cdn.jsdelivr.net/npm/html2canvas@1.4.1/dist/html2canvas.min.js"></script> <link
rel="stylesheet" href="styles.css"> <link rel="stylesheet" href="login.css"> <script>if('serviceWorker' in
navigator){navigator.serviceWorker.register('sw.js')}</script> </head> <body> <!-- HEADER --> <header
class="header"> <div class="header-left">  <div>
<div class="logo-stack"> <div class="station-logo-crop" style="width:45px;height:45px;"> <svg viewBox="0 0
100 100" style="width:100%;height:100%;"> <defs> <linearGradient id="shieldGrad2" x1="0" y1="0" x2="1"
y2="1"> <stop offset="0%" stop-color="#00f0ff" /> <stop offset="100%" stop-color="#00e676" />
</linearGradient> </defs> <path d="M50,5 L90,25 L90,75 L50,95 L10,75 L10,25 Z" fill="none"
stroke="url(#shieldGrad2)" stroke-width="3" /> <rect x="40" y="45" width="8" height="35" fill="#00f0ff"
opacity="0.8" /> <rect x="52" y="35" width="8" height="45" fill="#00e676" opacity="0.8" /> <path d="M30,40
Q50,20 70,40" fill="none" stroke="#ffc107" stroke-width="2" /> <circle cx="50" cy="55" r="10" fill="#1a2030"
stroke="#00f0ff" stroke-width="1" /> <path d="M50,50 Q55,55 50,62 Q45,55 50,50" fill="#ffab40" /> </svg>
</div> <h1 class="header-title">■■ Hammar IPF - Smart SCADA</h1> </div> <p class="header-sub">Advanced GOSP
Simulation &amp; Engineering Calculator – API/ASTM Standards</p> </div> <div class="header-right">
<button class="icon-btn" id="voiceBtn" onclick="toggleVoice()" title="Voice Control">■</button> <button
class="icon-btn" id="themeBtn" onclick="toggleTheme()" title="Toggle Theme">■</button> <!-- ALARMS DROPDOWN
--> <div class="alarm-dropdown-container"> <button class="icon-btn" id="alarmToggle"
onclick="toggleAlarms()" title="System Alarms">
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■ <span id="alarmCount" class="badge-mini">0</span> </button> <div class="alarm-panel minimized"
id="alarmPanel"> <div class="alarm-head" onclick="toggleAlarms()"> ■ <span data-i18n="alarms_title">System
Alarms</span> </div> <div class="alarm-list" id="alarmList"></div> </div> <button class="sim-btn"
id="simBtn" onclick="toggleSimulation()" data-i18n="sim_off">■■ Live: OFF</button> <button class="lang-btn"
id="langBtn" onclick="toggleLang()">■■■■■■■■</button> <div class="pulse-dot"></div><span class="sys-on"
data-i18n="status_online">ONLINE</span> <span class="clock" id="clock"></span> </div> <!-- LAYOUT
--> <div class="layout"> <!-- SIDEBAR NAV --> <nav class="sidebar" id="sidebar"> <button class="nav-btn
active" data-tab="overview"> <svg class="nav-icon" viewBox="0 0 24 24"><path
d="M10,20V14H14V20H19V12H22L12,3L2,12H5V20H10Z" fill="currentColor"/></svg> Overview </button> <button
class="nav-btn" data-tab="separator"> <svg class="nav-icon" viewBox="0 0 24 24"><path d="M12,2A10,10 0 0,0
2,12A10,10 0 0,0 12,22A10,10 0 0,0 22,12A10,10 0 0,0 12,2M12,4A8,8 0 0,1 20,12A8,8 0 0,1 12,20A8,8 0 0,1
4,12A8,8 0 0,1 12,4M12,6A6,6 0 0,0 6,12A6,6 0 0,0 12,18A6,6 0 0,0 18,12A6,6 0 0,0 12,6Z"
fill="currentColor"/></svg> Separator Calc </button> <button class="nav-btn" data-tab="trains"> <svg
class="nav-icon" viewBox="0 0 24 24"><path d="M12,2L4.5,20.29L5.21,21L12,18L18.79,21L19.5,20.29L12,2Z"
fill="currentColor"/></svg> Production Trains </button> <button class="nav-btn" data-tab="gas"> <svg
class="nav-icon" viewBox="0 0 24 24"><path d="M17.66,11.2C17.43,10.9 17.15,10.64 16.89,10.38C16.22,9.7
15.46,9.03 15.46,7.73C15.46,6.5 15.93,5.17 16.12,4.04C16.17,3.77 16.22,3.5 16.22,3.22C16.22,2.56 15.93,2
15.43,1.72C14.77,1.35 14.05,1.88 13.79,2.54L13.77,2.56C13.29,3.77 12.63,4.94 12.63,6.23V6.24C12.63,7.74
13.57,9.05 13.57,10.56C13.57,11.3 13.04,11.9 12.47,12.46C11.66,13.28 10.8,14.11 10.8,15.91C10.8,18.9
13.3,21.32 16.46,21.32C19.61,21.32 22.11,18.9 22.11,15.91C22.11,14.3 21.11,12.91
20.11,11.83L17.66,11.2M14.6,20.08C13.85,20.08 13.25,19.46 13.25,18.71C13.25,17.96 13.85,17.34
14.6,17.34C15.35,17.34 15.95,17.96 15.95,18.71C15.95,19.46 15.35,20.08 14.6,20.08M16.63,16.03C15.38,16.03
14.38,15.03 14.38,13.78C14.38,12.53 15.38,11.53 16.63,11.53C17.88,11.53 18.88,12.53 18.88,13.78C18.88,15.03
17.88,16.03 16.63,16.03Z" fill="currentColor"/></svg> Gas Treatment </button> <button class="nav-btn"
data-tab="oil"> <svg class="nav-icon" viewBox="0 0 24 24"><path d="M12,2C12,2 6,10 6,14A6,6 0 0,0 12,20A6,6
0 0,0 18,14C18,10 12,2 12,2Z" fill="currentColor"/></svg> Oil Treatment </button> <button class="nav-btn"
data-tab="water"> <svg class="nav-icon" viewBox="0 0 24 24"><path d="M12,20A6,6 0 0,1 6,14C6,10 12,3.25
12,3.25C12,3.25 18,10 18,14A6,6 0 0,1 12,20Z" fill="currentColor"/></svg> Water Treatment </button> <button
class="nav-btn" data-tab="chemical"> <svg class="nav-icon" viewBox="0 0 24 24"><path
d="M12,2L4.5,20.29L5.21,21L12,18L18.79,21L19.5,20.29L12,2Z" fill="currentColor"/></svg> Chemical Injection
</button> <button class="nav-btn" data-tab="storage"> <svg class="nav-icon" viewBox="0 0 24 24"><path d="M18
,18H6V17H18V18M18,16H6V15H18V16M18,14H6V13H18V14M18,12H6V11H18V12M18,10H6V9H18V10M18,8H6V7H18V8M19,4H5A2,2 0
0,0 3,6V18A2,2 0 0,0 5,20H19A2,2 0 0,0 21,18V6A2,2 0 0,0 19,4Z" fill="currentColor"/></svg> Storage & Export
</button> <button class="nav-btn" data-tab="charts"> <svg class="nav-icon" viewBox="0 0 24 24"><path
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d="M16,11.78L20.24,4.45L21.97,5.45L16.74,14.5L10.23,10.75L5.46,19H22V21H2V3H4V17.54L9.5,8L16,11.78Z"
fill="currentColor"/></svg> Charts </button> <button class="nav-btn" data-tab="equations"> <svg
class="nav-icon" viewBox="0 0 24 24"><path d="M19,3H5C3.89,3 3,3.89 3,5V19C3,20.11 3.89,21 5,21H19C20.11,21
21,20.11 21,19V5C21,3.89 20.11,3 19,3M19,19H5V5H19V19M11,7H13V9H11V7M11,11H13V13H11V11M11,15H13V17H11V15M7,7
H9V9H7V7M7,11H9V13H7V11M7,15H9V17H7V15M15,7H17V9H15V7M15,11H17V13H15V11M15,15H17V17H15V15Z"
fill="currentColor"/></svg>
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Equations </button> <button class="nav-btn" data-tab="diagram"> <svg class="nav-icon" viewBox="0 0 24
24"><path d="M9.5,3A6.5,6.5 0 0,1 16,9.5C16,11.11 15.41,12.59
14.44,13.73L14.71,14H15.5L20.5,19L19,20.5L14,15.5V14.71L13.73,14.44C12.59,15.41 11.11,16 9.5,16A6.5,6.5 0
0,1 3,9.5A6.5,6.5 0 0,1 9.5,3M9.5,5C7,5 5,7 5,9.5C5,12 7,14 9.5,14C12,14 14,12 14,9.5C14,7 12,5 9.5,5Z"
fill="currentColor"/></svg> Process Diagram </button> <button class="nav-btn" data-tab="control"> <svg
class="nav-icon" viewBox="0 0 24 24"><path d="M21,16.5C21,16.88 20.79,17.21
20.47,17.38L12.57,21.82C12.41,21.94 12.21,22 12,22C11.79,22 11.59,21.94 11.43,21.82L3.53,17.38C3.21,17.21
3,16.88 3,16.5V7.5C3,7.12 3.21,6.79 3.53,6.62L11.43,2.18C11.59,2.06 11.79,2 12,2C12.21,2 12.41,2.06
12.57,2.18L20.47,6.62C20.79,6.79 21,7.12 21,7.5V16.5Z" fill="currentColor"/></svg> Control Panel </button>
</nav> <!-- MAIN CONTENT --> <main class="content" id="content"> <!-- ===== OVERVIEW TAB ===== --> <div
class="tab-panel active" id="tab-overview"> <div class="panel-nav"><button class="nav-arrow"
onclick="histBack()" title="Back"><svg width="18" height="18" viewBox="0 0 24 24" fill="none"
stroke="currentColor" stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="15
18 9 12 15 6"/></svg></button><button class="nav-arrow" onclick="histFwd()" title="Forward"><svg width="18"
height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="9 18 15 12 9 6"/></svg></button><button class="save-btn"
onclick="exportPDF()" style="margin-right:8px;background:#e91e63;color:#fff;border:none;">■ Daily
Report</button><button class="save-btn" onclick="savePanel('tab-overview')">■ Save</button></div> <div
class="panel-title" data-i18n="overview_title">Plant Overview – Process Flow Schematic</div> <!-- Live
System Summary Panel --> <div class="calc-card glass" style="margin-bottom:20px; border-left: 4px solid
var(--gas-light);"> <div style="display:flex; justify-content:space-between; align-items:center;
margin-bottom:15px;"> <h3 style="font-size:14px; color:var(--text); font-weight:800;
text-transform:uppercase; letter-spacing:1px;">■ Live System Summary</h3> <div style="display:flex;
gap:10px;"> <span class="sys-on" style="background:var(--gas-bg); padding:4px 10px; border-radius:4px;
font-size:10px;">GOSP-1 ACTIVE</span> <span class="sys-on" style="background:var(--water-bg);
color:var(--water); padding:4px 10px; border-radius:4px; font-size:10px;">TUNNEL SECURED</span> </div>
</div> <div style="display:grid; grid-template-columns: repeat(auto-fit, minmax(150px, 1fr)); gap:15px;">
<div style="background:rgba(0,0,0,0.02); padding:10px; border-radius:8px; border:1px solid
var(--card-border);"> <div style="font-size:10px; color:var(--text3); text-transform:uppercase;">Station
Throughput</div> <div style="font-family:var(--mono); font-size:18px; font-weight:700;
color:var(--oil-light);" id="sum-inlet">135,420 <span style="font-size:10px;">bpd</span></div> </div> <div
style="background:rgba(0,0,0,0.02); padding:10px; border-radius:8px; border:1px solid var(--card-border);">
<div style="font-size:10px; color:var(--text3); text-transform:uppercase;">Export Quality (BS&W)</div> <div
style="font-family:var(--mono); font-size:18px; font-weight:700; color:var(--gas-light);">0.15 <span
style="font-size:10px;">%</span></div> </div> <div style="background:rgba(0,0,0,0.02); padding:10px;
border-radius:8px; border:1px solid var(--card-border);"> <div style="font-size:10px; color:var(--text3);
text-transform:uppercase;">Gas Energy Output</div> <div style="font-family:var(--mono); font-size:18px;
font-weight:700; color:var(--water-light);">12.4 <span style="font-size:10px;">MMBTU/D</span></div> </div>
<div style="background:rgba(0,0,0,0.02); padding:10px; border-radius:8px; border:1px solid
var(--card-border);"> <div style="font-size:10px; color:var(--text3); text-transform:uppercase;">System
Health</div> <div style="font-family:var(--mono); font-size:18px; font-weight:700;
color:var(--gas);">STABLE</div> </div> </div> <!-- SVG Process Schematic --> <div
class="schematic-wrap"> <svg id="plantSVG" viewBox="0 0 1100 520" xmlns="http://www.w3.org/2000/svg"> <defs>
<linearGradient id="oilG" x1="0" y1="0" x2="1" y2="0"><stop offset="0%" stop-color="#4a2800"/><stop
offset="100%" stop-color="#a05a00"/></linearGradient> <linearGradient id="gasG" x1="0" y1="0" x2="1"
y2="0"><stop offset="0%" stop-color="#8b0000"/><stop offset="100%" stop-color="#ff4444"/></linearGradient>
<linearGradient id="watG" x1="0" y1="0" x2="1" y2="0"><stop offset="0%" stop-color="#003366"/><stop
offset="100%" stop-color="#0088cc"/></linearGradient> <linearGradient id="vesG" x1="0" y1="0" x2="0"
y2="1"><stop offset="0%" stop-color="#2a3a5a"/><stop offset="100%" stop-color="#1a2540"/></linearGradient>
<filter id="glw"><feGaussianBlur stdDeviation="2" result="blur"/><feMerge><feMergeNode
in="blur"/><feMergeNode in="SourceGraphic"/></feMerge></filter> <marker id="arro" viewBox="0 0 10 6"
refX="10" refY="3" markerWidth="8" markerHeight="6" orient="auto"><path d="M0,0 L10,3 L0,6z"
fill="#a05a00"/></marker>
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<marker id="arrG" viewBox="0 0 10 6" refX="10" refY="3" markerWidth="8" markerHeight="6" orient="auto"><path
d="M0,0 L10,3 L0,6z" fill="#ff4444"/></marker> <marker id="arrW" viewBox="0 0 10 6" refX="10" refY="3"
markerWidth="8" markerHeight="6" orient="auto"><path d="M0,0 L10,3 L0,6z" fill="#0088cc"/></marker> <marker
id="arrC" viewBox="0 0 10 6" refX="10" refY="3" markerWidth="8" markerHeight="6" orient="auto"><path d="M0,0
L10,3 L0,6z" fill="#00e676"/></marker> </defs> <!-- Background grid --> <pattern id="grid" width="20"
height="20" patternUnits="userSpaceOnUse"><path d="M 20 0 L 0 0 20" fill="none"
stroke="rgba(0,240,255,0.03)" stroke-width="0.5"/></pattern> <rect width="1100" height="520"
fill="url(#grid)"/> <!-- WELLHEAD --> <g class="svg-node" onclick="switchTab('separator')"
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style="cursor:pointer"> <rect x="20" y="180" width="80" height="100" rx="6" fill="url(#vesG)"
stroke="#00f0ff" stroke-width="1"/> <line x1="60" y1="180" x2="60" y2="140" stroke="#00f0ff"
stroke-width="2"/> <circle cx="60" cy="135" r="8" fill="none" stroke="#00f0ff" stroke-width="1.5"/> <line
x1="45" y1="135" x2="75" y2="135" stroke="#00f0ff" stroke-width="1.5"/> <text x="60" y="240"
text-anchor="middle" fill="#00f0ff" font-size="10" font-weight="700"
data-i18n="svg_wellhead">WELLHEAD</text> <text x="60" y="255" text-anchor="middle" fill="#5a7aaa"
font-size="9" id="sv-well">- bpd</text> </g> <!-- Flow: Wellhead to Manifold --> <line x1="100" y1="230"
x2="150" y2="230" stroke="url(#oilG)" stroke-width="3" marker-end="url(#arro)"
stroke-dasharray="8,4"><animate attributeName="stroke-dashoffset" from="12" to="0" dur="1s"
repeatCount="indefinite"/></line> <!-- MANIFOLD --> <g class="svg-node"> <rect x="155" y="200" width="70"
height="60" rx="6" fill="url(#vesG)" stroke="#ffab40" stroke-width="1"/> <line x1="175" y1="215" x2="175"
y2="245" stroke="#ffab40" stroke-width="1.5"/> <line x1="190" y1="215" x2="190" y2="245" stroke="#ffab40"
stroke-width="1.5"/> <line x1="205" y1="215" x2="205" y2="245" stroke="#ffab40" stroke-width="1.5"/> <text
x="190" y="275" text-anchor="middle" fill="#ffab40" font-size="10" font-weight="700"
data-i18n="svg_manifold">MANIFOLD</text> <text x="190" y="290" text-anchor="middle" fill="#5a7aaa"
font-size="9" id="sv-manifold">- bpd</text> </g> <!-- Flow: Manifold to HP Sep --> <line x1="225" y1="230"
x2="280" y2="230" stroke="url(#oilG)" stroke-width="3" marker-end="url(#arro)"
stroke-dasharray="8,4"><animate attributeName="stroke-dashoffset" from="12" to="0" dur="1s"
repeatCount="indefinite"/></line> <!-- HP SEPARATOR (horizontal vessel) --> <g class="svg-node"
onclick="switchTab('separator')" style="cursor:pointer"> <rect x="285" y="195" width="140" height="70"
rx="35" fill="url(#vesG)" stroke="#00e676" stroke-width="1.5"/> <line x1="285" y1="235" x2="425" y2="235"
stroke="#a05a00" stroke-width="1" opacity="0.5"/> <rect x="310" y="210" width="90" height="15" rx="3"
fill="#a05a00" opacity="0.3"/> <rect x="310" y="235" width="90" height="15" rx="3" fill="#003366"
opacity="0.3"/> <text x="355" y="275" text-anchor="middle" fill="#00e676" font-size="10" font-weight="700"
data-i18n="svg_hpsep">HP SEPARATOR</text> <text x="355" y="290" text-anchor="middle" fill="#5a7aaa"
font-size="9" id="sv-hpsep">- bpd</text> </g> <!-- Flow: HP to LP --> <line x1="425" y1="230" x2="480"
y2="230" stroke="url(#oilG)" stroke-width="3" marker-end="url(#arro)" stroke-dasharray="8,4"><animate
attributeName="stroke-dashoffset" from="12" to="0" dur="1s" repeatCount="indefinite"/></line> <!-- LP
SEPARATOR --> <g class="svg-node" onclick="switchTab('separator')" style="cursor:pointer"> <rect x="485"
y="195" width="130" height="70" rx="35" fill="url(#vesG)" stroke="#00e676" stroke-width="1.5"/> <rect
x="505" y="210" width="85" height="15" rx="3" fill="#a05a00" opacity="0.3"/> <rect x="505" y="235"
width="85" height="15" rx="3" fill="#003366" opacity="0.3"/> <text x="550" y="275" text-anchor="middle"
fill="#00e676" font-size="10" font-weight="700" data-i18n="svg_lpsep">LP SEPARATOR</text> <text x="550"
y="290" text-anchor="middle" fill="#5a7aaa" font-size="9" id="sv-lpsep">- bpd</text> </g> <!-- GAS out from
HP Sep --> <line x1="355" y1="195" x2="355" y2="80" stroke="url(#gasG)" stroke-width="2"
marker-end="url(#arrG)" stroke-dasharray="6,3"><animate attributeName="stroke-dashoffset" from="9" to="0"
dur="0.8s" repeatCount="indefinite"/></line> <!-- GAS TREATMENT --> <g class="svg-node"
onclick="switchTab('gas')" style="cursor:pointer"> <rect x="300" y="30" width="110" height="50" rx="8"
fill="url(#vesG)" stroke="#ff4444" stroke-width="1.5"/> <polygon points="340,42 340,68 360,55" fill="none"
stroke="#ff4444" stroke-width="1.5"/> <circle cx="380" cy="55" r="10" fill="none" stroke="#ff4444"
stroke-width="1"/> <text x="355" y="95" text-anchor="middle" fill="#ff4444" font-size="10" font-weight="700"
data-i18n="svg_gas">GAS TREATMENT</text> <text x="355" y="110" text-anchor="middle" fill="#5a7aaa"
font-size="9" id="sv-gas">- MMSCFD</text> </g> <!-- Gas export line --> <line x1="410" y1="55" x2="500"
y2="55" stroke="url(#gasG)" stroke-width="2" marker-end="url(#arrG)"/>

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<text x="500" y="50" fill="#ff4444" font-size="9" font-weight="600" data-i18n="svg_gasexp">GAS EXPORT</text>
<!-- OIL out from LP Sep --> <line x1="615" y1="230" x2="670" y2="230" stroke="url(#oilG)" stroke-width="3"
marker-end="url(#arro)" stroke-dasharray="8,4"><animate attributeName="stroke-dashoffset" from="12" to="0"
dur="1s" repeatCount="indefinite"/></line> <!-- OIL TREATMENT --> <g class="svg-node"
onclick="switchTab('oil')" style="cursor:pointer"> <rect x="675" y="200" width="120" height="60" rx="8"
fill="url(#vesG)" stroke="#ff9800" stroke-width="1.5"/> <rect x="690" y="212" width="35" height="36" rx="4"
fill="none" stroke="#ff9800" stroke-width="1" stroke-dasharray="3,2"/> <text x="707" y="234"
text-anchor="middle" fill="#ff9800" font-size="8">HTR</text> <rect x="740" y="212" width="40" height="36"
rx="4" fill="none" stroke="#ff9800" stroke-width="1"/> <text x="760" y="234" text-anchor="middle"
fill="#ff9800" font-size="8">DESAL</text> <text x="735" y="275" text-anchor="middle" fill="#ff9800"
font-size="10" font-weight="700" data-i18n="svg_oil">OIL TREATMENT</text> <text x="735" y="290"
text-anchor="middle" fill="#5a7aaa" font-size="9" id="sv-oil">- bpd</text> </g> <!-- WATER out from LP Sep
--> <line x1="550" y1="265" x2="550" y2="370" stroke="url(#watG)" stroke-width="2" marker-end="url(#arrW)"
stroke-dasharray="6,3"><animate attributeName="stroke-dashoffset" from="9" to="0" dur="0.8s"
repeatCount="indefinite"/></line> <!-- WATER TREATMENT --> <g class="svg-node" onclick="switchTab('water')"
style="cursor:pointer"> <rect x="490" y="375" width="120" height="55" rx="8" fill="url(#vesG)"
stroke="#0088cc" stroke-width="1.5"/> <ellipse cx="530" cy="402" rx="18" ry="14" fill="none"
stroke="#0088cc" stroke-width="1"/> <ellipse cx="570" cy="402" rx="18" ry="14" fill="none" stroke="#0088cc"
stroke-width="1"/> <text x="550" y="445" text-anchor="middle" fill="#0088cc" font-size="10"
font-weight="700" data-i18n="svg_water">WATER TREATMENT</text> <text x="550" y="460" text-anchor="middle"
fill="#5a7aaa" font-size="9" id="sv-water">- bpd</text> </g> <!-- Oil to Storage --> <line x1="795" y1="230"
x2="850" y2="230" stroke="url(#oilG)" stroke-width="3" marker-end="url(#arro)"
stroke-dasharray="8,4"><animate attributeName="stroke-dashoffset" from="12" to="0" dur="1s"
repeatCount="indefinite"/></line> <!-- STORAGE TANKS --> <g class="svg-node" onclick="switchTab('storage')"

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style="cursor:pointer"> <rect x="855" y="185" width="50" height="80" rx="4" fill="url(#vesG)"
stroke="#ffc107" stroke-width="1.5"/> <rect x="860" y="220" width="40" height="40" rx="2" fill="#a05a00"
opacity="0.25"/> <rect x="915" y="185" width="50" height="80" rx="4" fill="url(#vesG)" stroke="#ffc107"
stroke-width="1.5"/> <rect x="920" y="210" width="40" height="50" rx="2" fill="#a05a00" opacity="0.3"/>
<text x="910" y="280" text-anchor="middle" fill="#ffc107" font-size="10" font-weight="700"
data-i18n="svg_storage">STORAGE</text> <text x="910" y="295" text-anchor="middle" fill="#5a7aaa"
font-size="9" id="sv-storage">— bbl</text> </g> <!-- Export --> <line x1="965" y1="225" x2="1020" y2="225"
stroke="url(#oilG)" stroke-width="3" marker-end="url(#arr0)"/> <g class="svg-node"> <rect x="1020" y="200"
width="65" height="50" rx="8" fill="url(#vesG)" stroke="#00e676" stroke-width="1.5"/> <text x="1052" y="230"
text-anchor="middle" fill="#00e676" font-size="16">■</text> <text x="1052" y="265" text-anchor="middle"
fill="#00e676" font-size="10" font-weight="700" data-i18n="svg_export">EXPORT</text> <text x="1052" y="280"
text-anchor="middle" fill="#5a7aaa" font-size="9" id="sv-export">— bpd</text> </g> <!-- CHEMICAL INJECTION
--> <g class="svg-node" onclick="switchTab('chemical')" style="cursor:pointer"> <rect x="670" y="380"
width="110" height="45" rx="8" fill="url(#vesG)" stroke="#e91e63" stroke-width="1"/> <text x="725" y="407"
text-anchor="middle" fill="#e91e63" font-size="9" font-weight="700" data-i18n="svg_chem">CHEMICAL
INJ.</text> <line x1="725" y1="380" x2="725" y2="260" stroke="#e91e63" stroke-width="1"
stroke-dasharray="4,3"><animate attributeName="stroke-dashoffset" from="7" to="0" dur="1.2s"
repeatCount="indefinite"/></line> <circle cx="725" cy="340" r="4" fill="#e91e63" opacity="0.6"><animate
attributeName="cy" from="380" to="260" dur="2s" repeatCount="indefinite"/></circle> </g> <!-- Legend --> <g
transform="translate(20,470)"> <rect width="14" height="8" rx="2" fill="#a05a00"/><text x="20" y="8"
fill="#7a8aaa" font-size="9" data-i18n="svg_leg_oil">Oil Flow</text> <rect x="90" width="14" height="8"
rx="2" fill="#ff4444"/><text x="110" y="8" fill="#7a8aaa" font-size="9" data-i18n="svg_leg_gas">Gas
Flow</text> <rect x="180" width="14" height="8" rx="2" fill="#0088cc"/><text x="200" y="8" fill="#7a8aaa"
font-size="9" data-i18n="svg_leg_water">Water Flow</text> <rect x="290" width="14" height="8" rx="2"
fill="#e91e63"/><text x="310" y="8" fill="#7a8aaa" font-size="9" data-i18n="svg_leg_chem">Chemical
Inj.</text> </g> </svg> </div> <!-- KPI Summary --> <div class="kpi-row" id="overviewKPI"></div>

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</div> <!-- ===== SEPARATOR CALCULATOR TAB ===== --> <div class="tab-panel" id="tab-separator"> <div
class="panel-nav"><button class="nav-arrow" onclick="histBack()"><svg width="18" height="18" viewBox="0 0 24
24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="15 18 9 12 15 6"/></svg></button><button class="nav-arrow"
onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="9 18 15 12 9
6"/></svg></button><button class="save-btn" onclick="savePanel('tab-separator')">■ Save</button></div> <div
class="panel-title">3-Phase Separator Engineering Calculator</div> <p class="panel-desc">API 12J / ASTM
standards – Souders-Brown &amp; Stokes' Law</p> <div class="calc-grid"> <!-- Inputs --> <div
class="calc-card"> <div class="card-head blue">Operating Conditions</div> <div class="input-group">
<label>Operating Pressure<input type="number" id="sep-P" value="150" step="5"> <span>psig</span></label>
<label>Operating Temperature<input type="number" id="sep-T" value="120" step="1"> <span>°F</span></label>
<label>Oil Flow Rate<input type="number" id="sep-Qo" value="25000" step="100"> <span>bpd</span></label>
<label>Water Flow Rate<input type="number" id="sep-Qw" value="10000" step="100"> <span>bpd</span></label>
<label>Gas Flow Rate<input type="number" id="sep-Qg" value="50" step="0.5"> <span>MMSCFD</span></label>
<label>Oil API Gravity<input type="number" id="sep-API" value="32" step="0.5"> <span>°API</span></label>
<label>Gas Specific Gravity<input type="number" id="sep-SGg" value="0.75" step="0.01">
<span></span></label> <label>Oil Viscosity<input type="number" id="sep-mu0" value="3.5" step="0.1">
<span>cp</span></label> <label>Droplet Diameter<input type="number" id="sep-dp" value="150" step="10">
<span>micron</span></label> <label>K Factor (Souders-Brown)<input type="number" id="sep-K" value="0.35"
step="0.01"> <span></span></label> <label>Retention Time (Oil)<input type="number" id="sep-tr0" value="3"
step="0.5"> <span>min</span></label> <label>Retention Time (Water)<input type="number" id="sep-trw"
value="3" step="0.5"> <span>min</span></label> </div> <button class="calc-btn" onclick="calcSeparator()">■
Calculate Separator</button> </div> <!-- Results --> <div class="calc-card"> <div class="card-head
green">Separator Results</div> <div id="sepResults" class="results-grid"></div> <div class="card-head green"
style="margin-top:16px">Vessel Specification</div> <div id="sepVessel" class="results-grid"></div> </div>
</div> <div class="chart-box"><canvas id="chartSep"></canvas></div> <!-- ===== PRODUCTION TRAINS TAB
===== --> <div class="tab-panel" id="tab-trains"> <div class="panel-nav"><button class="nav-arrow"
onclick="histBack()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="15 18 9 12 15
6"/></svg></button><button class="nav-arrow" onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24
24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="9 18 15 12 9 6"/></svg></button><button class="save-btn"
onclick="savePanel('tab-trains')">■ Save</button></div> <div class="panel-title">Production Trains – Live
Data</div> <div class="table-wrap"> <table class="dtable" id="trainInputTable">
<thead><tr><th>Train</th><th>Status</th><th>Inlet (bpd)</th><th>Oil (bpd)</th><th>Water (bpd)</th><th>Gas
(MMSCFD)</th><th>Sep Vol (m³)</th></tr></thead> <tbody id="trainBody"></tbody> </table> </div> <div
class="panel-title" style="margin-top:20px">Calculated Results</div> <div class="table-wrap"> <table
class="dtable res"> <thead><tr><th>Train</th><th>Water Cut %</th><th>GOR (SCF/bbl)</th><th>Inlet
(m³/d)</th><th>Retention (min)</th></tr></thead> <tbody id="trainResults"></tbody> </table> </div> <div
class="kpi-row" id="trainKPI"></div> </div> <!-- ===== GAS TREATMENT TAB ===== --> <div class="tab-panel"
id="tab-gas">

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<div class="panel-nav"><button class="nav-arrow" onclick="histBack()"><svg width="18" height="18" viewBox="0
0 24 24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="15 18 9 12 15 6"/></svg></button><button class="nav-arrow"
onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="9 18 15 12 9
6"/></svg></button><button class="save-btn" onclick="savePanel('tab-gas')">■ Save</button></div> <div
class="panel-title">Gas Treatment & Processing</div> <p class="panel-desc">Dehydration, sweetening,
compression – per GPSA/API standards</p> <div class="calc-grid"> <div class="calc-card"> <div
class="card-head orange">Gas Inlet Conditions</div> <div class="input-group"> <label>Gas Rate<input
type="number" id="gas-Q" value="50" step="1"> <span>MMSCFD</span></label> <label>Inlet Pressure<input
type="number" id="gas-Pin" value="150" step="5"> <span>psig</span></label> <label>Inlet Temperature<input
type="number" id="gas-Tin" value="120" step="1"> <span>°F</span></label> <label>Gas Sp. Gravity<input
type="number" id="gas-SG" value="0.75" step="0.01"> <span>–</span></label> <label>H2S Content<input
type="number" id="gas-H2S" value="2.5" step="0.1"> <span>mol%</span></label> <label>CO2 Content<input
type="number" id="gas-CO2" value="3.0" step="0.1"> <span>mol%</span></label> <label>Water Content<input
type="number" id="gas-H2O" value="50" step="1"> <span>lb/MMSCF</span></label> <label>Export Pressure<input
type="number" id="gas-Pout" value="1000" step="50"> <span>psig</span></label> </div> <button
class="calc-btn" onclick="calcGas()">■ Calculate Gas Treatment</button> </div> <div class="calc-card"> <div
class="card-head green">Gas Treatment Results</div> <div id="gasResults" class="results-grid"></div> </div>
</div> <div class="chart-box"><canvas id="chartGas"></canvas></div> </div> <!-- ===== OIL TREATMENT TAB
===== --> <div class="tab-panel" id="tab-oil"> <div class="panel-nav"><button class="nav-arrow"
onclick="histBack()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="15 18 9 12 15
6"/></svg></button><button class="nav-arrow" onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24
24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="9 18 15 12 9 6"/></svg></button><button class="save-btn"
onclick="savePanel('tab-oil')">■ Save</button></div> <div class="panel-title">Oil Treatment – Dehydration
& Desalting</div> <div class="calc-grid"> <div class="calc-card"> <div class="card-head purple">Oil
Treatment Input</div> <div class="input-group"> <label>Oil Rate<input type="number" id="oil-Q" value="25000"
step="100"> <span>bpd</span></label> <label>API Gravity<input type="number" id="oil-API" value="32"
step="0.5"> <span>°API</span></label> <label>Inlet BSW<input type="number" id="oil-BSWin" value="15"
step="0.5"> <span>%</span></label> <label>Target BSW<input type="number" id="oil-BSWout" value="0.5"
step="0.1"> <span>%</span></label> <label>Salt Content<input type="number" id="oil-salt" value="50"
step="5"> <span>PTB</span></label> <label>Target Salt<input type="number" id="oil-saltOut" value="10"
step="1"> <span>PTB</span></label> <label>Heater Temp In<input type="number" id="oil-Tin" value="100"
step="5"> <span>°F</span></label> <label>Heater Temp Out<input type="number" id="oil-Tout" value="150"
step="5"> <span>°F</span></label> <label>Wash Water Rate<input type="number" id="oil-wash" value="5"
step="0.5"> <span>% vol</span></label> </div> <button class="calc-btn" onclick="calcOil()">■ Calculate Oil
Treatment</button> </div> <div class="calc-card"> <div class="card-head green">Oil Treatment Results</div>
<div id="oilResults" class="results-grid"></div> </div> </div> <div class="chart-box"><canvas
id="chartOil"></canvas></div> </div> <!-- ===== WATER TREATMENT TAB ===== --> <div class="tab-panel"
id="tab-water"> <div class="panel-nav"><button class="nav-arrow" onclick="histBack()"><svg width="18"
height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="15 18 9 12 15 6"/></svg></button><button class="nav-arrow"
onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="9 18 15 12 9
6"/></svg></button><button class="save-btn" onclick="savePanel('tab-water')">■ Save</button></div> <div
class="panel-title">Produced Water Treatment</div> <div class="calc-grid"> <div class="calc-card">

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<div class="card-head cyan">Water Treatment Input</div> <div class="input-group"> <label>Water Rate<input
type="number" id="wat-Q" value="10000" step="100"> <span>bpd</span></label> <label>Oil in Water<input
type="number" id="wat-Oiw" value="500" step="10"> <span>ppm</span></label> <label>Target Oil in Water<input
type="number" id="wat-Oiwout" value="40" step="5"> <span>ppm</span></label> <label>TSS Inlet<input
type="number" id="wat-TSS" value="200" step="10"> <span>mg/L</span></label> <label>Temperature<input
type="number" id="wat-T" value="120" step="5"> <span>°F</span></label> <label>Skim Tank Retention<input
type="number" id="wat-ret" value="15" step="1"> <span>min</span></label> </div> <button class="calc-btn"
onclick="calcWater()">■ Calculate Water Treatment</button> </div> <div class="calc-card"> <div
class="card-head green">Water Treatment Results</div> <div id="waterResults" class="results-grid"></div>
</div> </div> <div class="chart-box"><canvas id="chartWater"></canvas></div> </div> <!-- ===== CHEMICAL
INJECTION TAB ===== --> <div class="tab-panel" id="tab-chemical"> <div class="panel-nav"><button
class="nav-arrow" onclick="histBack()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none"
stroke="currentColor" stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="15
18 9 12 15 6"/></svg></button><button class="nav-arrow" onclick="histFwd()"><svg width="18" height="18"
viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="9 18 15 12 9 6"/></svg></button><button class="save-btn"
onclick="savePanel('tab-chemical')">■ Save</button></div> <div class="panel-title">Chemical Injection
System</div> <div class="calc-grid"> <div class="calc-card"> <div class="card-head pink">Injection
Parameters</div> <div class="input-group"> <label>Total Liquid Rate<input type="number" id="chem-Ql"
value="35000" step="100"> <span>bpd</span></label> <label>Demulsifier Dose<input type="number"

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id="chem-demul" value="15" step="1"> <span>ppm</span></label> <label>Corrosion Inhibitor<input type="number"
id="chem-corr" value="10" step="1"> <span>ppm</span></label> <label>Scale Inhibitor<input type="number"
id="chem-scale" value="8" step="1"> <span>ppm</span></label> <label>H2S Scavenger<input type="number"
id="chem-h2s" value="20" step="1"> <span>ppm</span></label> <label>Biocide<input type="number" id="chem-bio"
value="50" step="5"> <span>ppm</span></label> <label>Oxygen Scavenger<input type="number" id="chem-o2"
value="5" step="1"> <span>ppm</span></label> <label>Pour Point Depressant<input type="number" id="chem-ppd"
value="100" step="10"> <span>ppm</span></label> </div> <button class="calc-btn" onclick="calcChemical()">■
Calculate Injection Rates</button> </div> <div class="calc-card"> <div class="card-head green">Chemical
Injection Results</div> <div id="chemResults" class="results-grid"></div> </div> </div> <div
class="chart-box"><canvas id="chartChem"></canvas></div> </div> <!-- ===== STORAGE & EXPORT TAB ===== -->
<div class="tab-panel" id="tab-storage"> <div class="panel-nav"><button class="nav-arrow"
onclick="histBack()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="15 18 9 12 15
6"/></svg></button><button class="nav-arrow" onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24
24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="9 18 15 12 9 6"/></svg></button><button class="save-btn"
onclick="savePanel('tab-storage')">■ Save</button></div> <div class="panel-title">Storage Tanks & Export</div> <div class="calc-grid"> <div class="calc-card"> <div class="card-head amber">Tank
Specifications</div> <div class="input-group"> <label>Number of Tanks<input type="number" id="stg-n"
value="4" step="1" min="1" max="20"> <span>units</span></label> <label>Tank Diameter<input type="number"
id="stg-D" value="60" step="1"> <span>m</span></label> <label>Tank Height<input type="number" id="stg-H"
value="18" step="0.5"> <span>m</span></label> <label>Max Fill Level<input type="number" id="stg-fill"
value="85" step="1"> <span>%</span></label> <label>Export Rate<input type="number" id="stg-exp"
value="100000" step="1000"> <span>bpd</span></label> <label>Oil Production<input type="number" id="stg-prod"
value="100000" step="1000"> <span>bpd</span></label>

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</div> <button class="calc-btn" onclick="calcStorage()">■ Calculate Storage</button> </div> <div
class="calc-card"> <div class="card-head green">Storage & Export Results</div> <div id="storageResults"
class="results-grid"></div> </div> </div> <!-- ===== CHARTS TAB ===== --> <div class="tab-panel"
id="tab-charts"> <div class="panel-nav"><button class="nav-arrow" onclick="histBack()"><svg width="18"
height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="15 18 9 12 15 6"/></svg></button><button class="nav-arrow"
onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="9 18 15 12 9
6"/></svg></button><button class="save-btn" onclick="savePanel('tab-charts')">■ Save</button></div> <div
class="panel-title">Engineering Charts & Analytics</div> <div class="charts-grid"> <div
class="chart-box"><div class="chart-title">Separator Efficiency vs Pressure</div><canvas
id="chartEffP"></canvas></div> <div class="chart-box"><div class="chart-title">GOR vs API
Gravity</div><canvas id="chartGorApi"></canvas></div> <div class="chart-box"><div
class="chart-title">Retention Time vs Flow Rate</div><canvas id="chartRetQ"></canvas></div> <div
class="chart-box"><div class="chart-title">Chemical Cost Breakdown</div><canvas
id="chartChemCost"></canvas></div> </div> <!-- ===== CONTROL PANEL TAB ===== --> <!-- ===== EQUATIONS
TAB ===== --> <div class="tab-panel" id="tab-equations"> <div class="panel-nav"><button class="nav-arrow"
onclick="histBack()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="15 18 9 12 15
6"/></svg></button><button class="nav-arrow" onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24
24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="9 18 15 12 9 6"/></svg></button><button class="save-btn"
onclick="savePanel('tab-equations')">■ Save</button></div> <div class="panel-title">■ Engineering Equations
– Manual Verification</div> <p class="panel-desc">API / ISO / ASTM / Darcy Standards – Enter values to verify
calculations</p> <div class="eq-grid" id="eqGrid"></div> </div> <!-- ===== PROCESS DIAGRAM DETAIL TAB =====
--> <div class="tab-panel" id="tab-diagram"> <div class="panel-nav"><button class="nav-arrow"
onclick="histBack()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="15 18 9 12 15
6"/></svg></button><button class="nav-arrow" onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24
24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="9 18 15 12 9 6"/></svg></button><button class="save-btn"
onclick="savePanel('tab-diagram')">■ Save</button></div> <div class="panel-title">■ Detailed Process Diagram
– Wellhead to Export</div> <p class="panel-desc">Click on any equipment to view details</p> <div
class="schematic-wrap" id="detailDiagramWrap"></div> <div id="equipPopup" class="svg-popup"></div> <!--
Detailed Process Schematic (Blueprint Style) --> <div class="calc-card glass" style="margin-top:20px;padding
:20px;overflow:hidden;height:auto;background:#061021;position:relative;border:1px solid
rgba(0,240,255,0.2);"> <div
style="position:absolute;top:15px;left:15px;background:rgba(0,180,255,0.1);padding:6px
12px;border-radius:6px;font-size:11px;color:#00f0ff;border:1px solid
rgba(0,240,255,0.3);z-index:2;font-weight:700;letter-spacing:1px;">DETAILED PROCESS SCHEMATIC – ENGINEERING
VIEW</div> <div class="blueprint-wrap"> <svg viewBox="0 0 1000 500" xmlns="http://www.w3.org/2000/svg"> <!--
Background Grid --> <defs> <pattern id="blueGrid" width="40" height="40" patternUnits="userSpaceOnUse">
<path d="M 40 0 L 0 0 0 40" fill="none" stroke="rgba(0,240,255,0.05)" stroke-width="0.5"/> </pattern>

```

```
</defs> <rect width="1000" height="500" fill="url(#blueGrid)" /> <!-- FLOW LINES --> <g
stroke="rgba(0,240,255,0.3)" stroke-width="2" fill="none"> <path d="M 50,250 L 150,250" /> <!-- Wellhead to
HP --> <path d="M 300,250 L 400,250" /> <!-- HP to LP --> <path d="M 530,250 L 630,250" /> <!-- LP to Treater
--> <path d="M 750,250 L 850,250" /> <!-- Treater to Storage --> <path d="M 225,210 V 100 H 350" /> <!-- HP
Gas out -->
```

```
<path d="M 465,290 V 400 H 550" /> <!-- LP Water out --> </g> <!-- EQUIPMENT --> <!-- Wellhead --> <rect
x="50" y="210" width="80" height="80" rx="4" fill="rgba(0,20,40,0.8)" stroke="#00f0ff" stroke-width="1.5" />
<text x="90" y="245" text-anchor="middle" fill="#00f0ff" font-size="10" font-weight="bold">WELLHEAD</text>
<text x="90" y="260" text-anchor="middle" fill="#5a7aaa" font-size="9">P: 2500 psi</text> <!-- HP SEPARATOR
--> <rect x="150" y="210" width="150" height="80" rx="40" fill="rgba(0,20,40,0.8)" stroke="#00e676"
stroke-width="1.5" /> <text x="225" y="245" text-anchor="middle" fill="#00e676" font-size="10"
font-weight="bold">HP SEPARATOR</text> <text x="225" y="260" text-anchor="middle" fill="#5a7aaa"
font-size="9">150 psig | 120°F</text> <!-- LP SEPARATOR --> <rect x="400" y="210" width="130" height="80"
rx="40" fill="rgba(0,20,40,0.8)" stroke="#00e676" stroke-width="1.5" /> <text x="465" y="245"
text-anchor="middle" fill="#00e676" font-size="10" font-weight="bold">LP SEPARATOR</text> <text x="465"
y="260" text-anchor="middle" fill="#5a7aaa" font-size="9">50 psig | 115°F</text> <!-- OIL TREATER --> <rect
x="630" y="210" width="120" height="80" rx="8" fill="rgba(0,20,40,0.8)" stroke="#ffab40" stroke-width="1.5"
/> <text x="690" y="245" text-anchor="middle" fill="#ffab40" font-size="10" font-weight="bold">OIL
TREATER</text> <text x="690" y="260" text-anchor="middle" fill="#5a7aaa"
font-size="9">Desalter/Heater</text> <!-- STORAGE --> <rect x="850" y="210" width="100" height="80" rx="4"
fill="rgba(0,20,40,0.8)" stroke="#ffc107" stroke-width="1.5" /> <text x="900" y="245" text-anchor="middle"
fill="#ffc107" font-size="10" font-weight="bold">STORAGE</text> <text x="900" y="260" text-anchor="middle"
fill="#5a7aaa" font-size="9">Export Tanks</text> <!-- GAS PLANT --> <rect x="350" y="70" width="120"
height="60" rx="8" fill="rgba(0,20,40,0.8)" stroke="#ff4444" stroke-width="1.5" /> <text x="410" y="100"
text-anchor="middle" fill="#ff4444" font-size="10" font-weight="bold">GAS PLANT</text> <text x="410" y="115"
text-anchor="middle" fill="#5a7aaa" font-size="9">Comp/Dehy</text> <!-- WATER TREATMENT --> <rect x="550"
y="370" width="120" height="60" rx="8" fill="rgba(0,20,40,0.8)" stroke="#0088cc" stroke-width="1.5" /> <text
x="610" y="400" text-anchor="middle" fill="#0088cc" font-size="10" font-weight="bold">WATER TREAT</text>
<text x="610" y="415" text-anchor="middle" fill="#5a7aaa" font-size="9">W0SEP/CPT</text> <!-- ARROWS -->
<marker id="arrowhead" markerWidth="10" markerHeight="7" refX="0" refY="3.5" orient="auto"> <polygon
points="0 0, 10 3.5, 0 7" fill="rgba(0,240,255,0.5)" /> </marker> </svg> </div> <div
style="position:absolute;bottom:0;left:0;right:0;background:linear-gradient(to top, rgba(0,0,0,0.8),
transparent);padding:30px;color:#fff;z-index:3;"> <h3 style="color:#00f0ff;margin:0 0
5px;font-size:18px;">Process Topology View</h3> <p
style="color:#5a7aaa;font-size:13px;margin:0;">Interactive schematic showing fluid flow and equipment
interconnections.</p> </div> </div> </div> <!-- ===== CONTROL PANEL TAB ===== --> <div class="tab-panel"
id="tab-control"> <div class="panel-nav"><button class="nav-arrow" onclick="histBack()"><svg width="18"
height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2.5" stroke-linecap="round"
stroke-linejoin="round"><polyline points="15 18 9 12 15 6"></svg></button><button class="nav-arrow"
onclick="histFwd()"><svg width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentColor"
stroke-width="2.5" stroke-linecap="round" stroke-linejoin="round"><polyline points="9 18 15 12 9 6"></svg></button></div> <div class="panel-title" data-i18n="nav_control">■ Master Control Panel</div> <!--
Role Selector -->
```

```
<div style="margin-bottom:15px;display:flex;gap:10px;align-items:center;"> <span
style="font-size:11px;color:var(--text3);">ACCESS LEVEL:</span> <select id="roleSelect" class="sts-select"
onchange="showToast('Access Level Changed to ' + this.value)"> <option value="Operator">Operator</option>
<option value="Engineer" selected>Engineer</option> <option value="Admin">Administrator</option> </select>
</div> <!-- Interactive PID Control Element --> <div class="pid-wrap"> <h3
style="color:#00e676;margin-bottom:10px;font-size:14px;" data-i18n="pid_title">PID Level Control – HP
Separator</h3> <div class="pid-flex"> <div class="pid-tank-wrap"> <svg viewBox="0 0 100 150" width="100"
height="150"> <rect x="10" y="10" width="80" height="130" rx="10" fill="none" stroke="#00f0ff"
stroke-width="2"/> <rect id="pidLevelRect" x="12" y="70" width="76" height="68" rx="8" fill="#a05a00"
opacity="0.8"/> <text id="pidLevelText" x="50" y="80" text-anchor="middle" fill="#fff" font-size="14"
font-weight="bold">50%</text> </svg> </div> <div class="pid-controls"> <label data-i18n="pid_valve">Liquid
Outflow Valve (%)</label> <input type="range" id="pidValveSlider" min="0" max="100" value="50"
oninput="updatePID()"> <div id="pidValveVal"
style="color:#00f0ff;font-weight:bold;margin-bottom:15px;">50%</div> <label
data-i18n="pid_retention">Retention Time (min)</label> <div id="pidRetentionVal"
style="color:#ffab40;font-size:18px;font-weight:bold;margin-bottom:15px;">3.00</div> <button
class="calc-btn" onclick="autoPID()" style="width:100%" data-i18n="pid_auto">Auto Target 50%</button> </div>
</div> <div class="control-grid" id="controlGrid"></div> <div class="success-banner"
data-i18n="success_msg">■ All systems are stable. Mass balance and chemical injection rates are
automatically updated based on live input.</div> </div> </main> <!-- ALARMS PANEL MOVED TO HEADER --> </div>
<div class="toast" id="toast"></div> <script src="login.js"></script> <script src="core.js"></script>
<script src="charts.js"></script> <script src="features.js"></script> </body> </html>
```

■ File: styles.css

```
*,*::before,*::after{box-sizing:border-box;margin:0;padding:0} html{scroll-behavior:smooth} body{font-family:
'Inter',system-ui,sans-serif;background:#f4f6f9;color:#1a2030;min-height:100vh;overflow-x:hidden} ::-webkit
-scrollbar{width:5px;height:5px}::-webkit-scrollbar-track{background:#e8ecf1}::-webkit-scrollbar-thumb{backg
round:#b0bcc8;border-radius:3px} @keyframes
fadeIn{from{opacity:0;transform:translateY(12px)}to{opacity:1;transform:none}} @keyframes
pulse{0%,100%{opacity:1}50%{opacity:.4}} @keyframes glow{0%,100%{box-shadow:0 0 8px
rgba(0,180,80,.2)}50%{box-shadow:0 0 20px rgba(0,180,80,.5)}} @keyframes
headerGradient{0%{background-position:0% 50%}50%{background-position:100% 50%}100%{background-position:0%
50%}} @keyframes arrowPulse{0%,100%{box-shadow:0 2px 8px rgba(0,0,0,.06)}50%{box-shadow:0 2px 12px
rgba(10,138,62,.15)}} /* ===== THREE-PHASE COLOR TOKENS ===== */ :root{ --oil: #4a2800; --oil-light:
#a05a00; --oil-bg: rgba(74, 40, 0, 0.08); --oil-border: rgba(74, 40, 0, 0.2); --gas: #00e676; --gas-light:
#69f0ae; --gas-bg: rgba(0, 230, 118, 0.08); --gas-border: rgba(0, 230, 118, 0.2); --water: #0088cc;
--water-light: #33b5e5; --water-bg: rgba(0, 136, 204, 0.08); --water-border: rgba(0, 136, 204, 0.2);
--accent: #00f0ff; --accent-glow: rgba(0, 240, 255, 0.4); --bg: #f8fafc; --card: rgba(255, 255, 255, 0.75);
--card-border: rgba(226, 232, 240, 0.8); --text: #0f172a; --text2: #475569; --text3: #94a3b8; --header-bg:
linear-gradient(135deg, #0f172a, #1e293b); --mono: 'JetBrains Mono', monospace; } body.dark-mode{ --bg:
#020617; --card: rgba(15, 23, 42, 0.8); --card-border: rgba(30, 41, 59, 0.5); --text: #f8fafc; --text2:
#94a3b8; --text3: #64748b; --card-shadow: 0 12px 40px rgba(0,0,0,0.6); }
.glass{backdrop-filter:blur(16px);-webkit-backdrop-filter:blur(16px);border:1px solid
var(--card-border);box-shadow:0 8px 32px rgba(0,0,0,0.05)} /* HEADER */
.header{background:linear-gradient(135deg,#0f1923,#162030,#1a2840,#162030,#0f1923);background-size:300%
300%;animation:headerGradient 10s ease infinite;border-bottom:2px solid var(--gas);padding:14px 24px;display
:flex;align-items:center;justify-content:space-between;position:sticky;top:0;z-index:100;backdrop-filter:blu
r(20px)} .header::after{content:'';position:absolute;bottom:0;left:0;right:0;height:3px;background:linear-gr
radient(90deg,var(--oil),var(--oil-light),var(--gas),var(--gas-light),var(--water),var(--water-light),var(--o
il));background-size:200% 100%;animation:headerGradient 6s linear infinite}
.header-left{display:flex;align-items:center;gap:12px} .header-title{font-size:22px;font-weight:900;backgrou
nd:linear-gradient(135deg,var(--accent),#fff,var(--gas-light));-webkit-background-clip:text;-webkit-text-fil
l-color:transparent;background-clip:text;letter-spacing:-0.5px} .header-sub{font-size:10px;color:rgba(255,25
5,255,0.6);margin-top:2px;letter-spacing:0.5px;text-transform:uppercase}
.header-right{display:flex;align-items:center;gap:10px}
.pulse-dot{width:7px;height:7px;border-radius:50%;background:var(--gas-light);animation:pulse 2s infinite}
.sys-on{font-family:var(--mono);font-size:9px;letter-spacing:.15em;color:var(--gas-light);font-weight:700}
.clock{font-family:var(--mono);font-size:10px;color:#8a9ab0;padding-left:10px;border-left:1px solid #3a4a60}
.lang-btn{background:rgba(255,255,255,.1);border:1px solid
rgba(255,255,255,.25);color:#fff;font-size:11px;font-weight:700;padding:6px
14px;border-radius:6px;cursor:pointer;font-family:'Inter',sans-serif;transition:all
.2s;letter-spacing:.03em} .lang-btn:hover{background:rgba(255,255,255,.2);border-color:#fff}
.icon-btn.active{background:var(--gas);border-color:var(--gas-light);box-shadow:0 0 15px var(--gas)} /* LOGO
STACK */ .logo-stack { display: flex; flex-direction: column; align-items: flex-start; gap: 4px; }
.station-logo-crop { height: 45px; width: auto; object-fit: cover; object-position: top; filter:
drop-shadow(0 0 8px rgba(0,240,255,0.4)); transition: transform 0.3s; } .station-logo-crop:hover {
transform: scale(1.05); } /* SCHEMATIC */
.schematic-wrap{background:linear-gradient(135deg,#fafcff,#f0f4f9);border:2px solid
transparent;border-image:linear-gradient(135deg,var(--oil-border),var(--gas-border),var(--water-border))
1;border-radius:14px;padding:20px;overflow-x:auto;margin-bottom:16px;box-shadow:0 4px 20px
rgba(0,0,0,.06),inset 0 1px 0 rgba(255,255,255,.8)}

.schematic-wrap svg{width:100%;height:auto;min-width:700px} .svg-node{transition:transform .2s}
.svg-node:hover{transform:scale(1.02)} .svg-node: hover rect, .svg-node: hover circle, .svg-node: hover
ellipse{filter:brightness(1.15) drop-shadow(0 0 6px rgba(0,230,118,.25));transition:filter .3s} .svg-node
text{font-family:var(--mono)} /* RTL Support */ html[dir="rtl"] .header-left{flex-direction:row-reverse}
html[dir="rtl"] .sidebar{border-right:none;border-left:1px solid var(--card-border)} html[dir="rtl"]
.nav-btn{text-align:right} html[dir="rtl"] .nav-btn.active{border-left:none;border-right:3px solid
var(--gas)} html[dir="rtl"] .clock{padding-left:0;padding-right:10px;border-left:none;border-right:1px solid
#3a4a60} html[dir="rtl"] .header-right{flex-direction:row-reverse} html[dir="rtl"]
.res-row{flex-direction:row-reverse} html[dir="rtl"] .input-group label{direction:rtl} html[dir="rtl"]
.dtable thead th,html[dir="rtl"] .dtable tbody td{text-align:left} html[dir="rtl"] .dtable thead
th:first-child,html[dir="rtl"] .dtable tbody td:first-child{text-align:right} html[dir="rtl"]
.panel-nav{flex-direction:row-reverse} /* LAYOUT */ .layout{display:flex;min-height:calc(100vh - 56px)} /*
SIDEBAR */ .sidebar{width:210px;background:#fff;border-right:1px solid var(--card-border);padding:12px
8px;display:flex;flex-direction:column;gap:3px;position:sticky;top:56px;height:calc(100vh -
56px);overflow-y:auto;box-shadow:2px 0 8px rgba(0,0,0,.03)}
.nav-btn{background:none;border:none;color:var(--text2);font-size:12px;font-weight:500;padding:10px
12px;border-radius:8px;text-align:left;cursor:pointer;transition:all
.2s;font-family:'Inter',sans-serif;display:flex;align-items:center;gap:10px}
```

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.nav-btn:hover{background:var(--gas-bg);color:var(--gas)}
.nav-btn.active{background:var(--gas-bg);color:var(--gas);border-left:3px solid var(--gas);font-weight:700}
.nav-icon{width:18px;height:18px;flex-shrink:0;transition:transform .2s} .nav-btn:hover
.nav-icon{transform:scale(1.1)} /* CONTENT */ .content{flex:1;padding:20px
28px;overflow-y:auto;max-height:calc(100vh - 56px);background:var(--bg)}
.tab-panel{display:none;animation:fadeIn .4s ease both} .tab-panel.active{display:block} .panel-title{font-s
ize:18px;font-weight:800;color:var(--text);margin-bottom:14px;padding-bottom:10px;border-bottom:2px solid
transparent;border-image:linear-gradient(90deg,var(--gas-border),transparent)
1;display:flex;align-items:center;gap:8px;letter-spacing:-.01em} /* PANEL NAVIGATION BAR (back/forward/save)
*/ .panel-nav{display:flex;align-items:center;gap:10px;margin-bottom:16px;padding:8px 0} .panel-nav
.nav-arrow{background:linear-gradient(135deg,#f8fafb,#fff);border:1.5px solid var(--card-border);color:var(-
--text2);width:40px;height:40px;border-radius:10px;display:flex;align-items:center;justify-content:center;cur
sor:pointer;transition:all .25s;box-shadow:0 2px 8px rgba(0,0,0,.06)} .panel-nav .nav-arrow
svg{width:18px;height:18px;transition:transform .2s} .panel-nav .nav-arrow:hover{background:linear-gradient(
135deg,var(--gas-bg),rgba(46,204,113,.08));color:var(--gas);border-color:var(--gas);box-shadow:0 4px 16px
rgba(10,138,62,.12);transform:translateY(-1px)} .panel-nav .nav-arrow:hover svg{transform:scale(1.15)}
.panel-nav .nav-arrow:active{transform:translateY(0);box-shadow:0 1px 4px rgba(0,0,0,.08)} .panel-nav
.nav-arrow:disabled{opacity:.3;cursor:not-allowed;transform:none} .panel-nav
.save-btn{background:linear-gradient(135deg,#f8fafb,#fff);border:1.5px solid
var(--card-border);color:var(--text2);height:40px;padding:0 18px;border-radius:10px;display:flex;align-items
:center;gap:6px;cursor:pointer;font-size:12px;font-weight:600;transition:all
.25s;font-family:'Inter',sans-serif;margin-left:auto;box-shadow:0 2px 8px rgba(0,0,0,.06)} .panel-nav .save-
btn:hover{background:linear-gradient(135deg,var(--gas-bg),rgba(46,204,113,.08));color:var(--gas);border-colo
r:var(--gas);box-shadow:0 4px 16px rgba(10,138,62,.12);transform:translateY(-1px)} html[dir="rtl"]
.panel-nav .save-btn{margin-left:0;margin-right:auto}
.panel-desc{font-size:12px;color:var(--text3);margin-bottom:16px} /* CALC GRID */
.calc-grid{display:grid;grid-template-columns:1fr 1fr;gap:20px;margin-bottom:20px}
.calc-card{background:var(--card);backdrop-filter:blur(16px);border:1.5px solid
var(--card-border);border-radius:16px;padding:22px;overflow:hidden;box-shadow:0 4px 16px
rgba(0,0,0,.04);transition:all .3s ease}
.calc-card:hover{border-color:var(--gas-border);transform:translateY(-4px);box-shadow:0 12px 32px
rgba(0,0,0,.08)}
.card-head{font-family:var(--mono);font-size:11px;letter-spacing:.12em;text-transform:uppercase;padding:8px
14px;margin:-20px -20px 16px 16px;font-weight:700}
.card-head.blue{background:var(--water-bg);color:var(--water);border-bottom:1px solid var(--water-border)}
.card-head.green{background:var(--gas-bg);color:var(--gas);border-bottom:1px solid var(--gas-border)}
.card-head.orange{background:rgba(255,152,0,.06);color:#e68a00;border-bottom:1px solid rgba(255,152,0,.15)}
.card-head.purple{background:rgba(156,39,176,.06);color:#9c27b0;border-bottom:1px solid
rgba(156,39,176,.15)}

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.card-head.cyan{background:var(--water-bg);color:var(--water);border-bottom:1px solid var(--water-border)}
.card-head.pink{background:rgba(233,30,99,.06);color:#c2185b;border-bottom:1px solid rgba(233,30,99,.15)}
.card-head.amber{background:rgba(255,160,0,.06);color:#e68a00;border-bottom:1px solid rgba(255,160,0,.15)}
.card-head.oil{background:var(--oil-bg);color:var(--oil);border-bottom:1px solid var(--oil-border)} /*
INPUTS */ .input-group{display:flex;flex-direction:column;gap:8px} .input-group
label{display:grid;grid-template-columns:1fr 100px
50px;align-items:center;gap:8px;font-size:12px;color:var(--text2)} .input-group
input{background:#f8fafb;border:1px solid
var(--card-border);border-radius:6px;color:var(--text);font-family:var(--mono);font-size:12px;padding:7px
10px;text-align:right;outline:none;transition:all .25s} .input-group
input:focus{border-color:var(--gas);box-shadow:0 0 3px rgba(10,138,62,.1)} .input-group
span{font-family:var(--mono);font-size:10px;color:var(--text3)} .calc-btn{width:100%;margin-top:14px;padding
:12px;background:linear-gradient(135deg,var(--gas),var(--gas-light));color:#fff;border:none;border-radius:8p
x;font-weight:700;font-size:13px;cursor:pointer;transition:all .2s;letter-spacing:.04em}
.calc-btn:hover{transform:translateY(-1px);box-shadow:0 4px 20px rgba(10,138,62,.3)}
.calc-btn:active{transform:translateY(0)} /* RESULTS GRID */
.results-grid{display:flex;flex-direction:column;gap:6px}
.res-row{display:flex;justify-content:space-between;align-items:center;padding:8px 0;border-bottom:1px solid
rgba(0,0,0,.04)} .res-row:last-child{border-bottom:none} .res-label{font-size:12px;color:var(--text2)}
.res-value{font-family:var(--mono);font-size:13px;font-weight:600;color:var(--text)}
.res-unit{font-family:var(--mono);font-size:10px;color:var(--text3);margin-left:4px} /* TABLES */
.table-wrap{overflow-x:auto;border-radius:12px;border:1px solid
var(--card-border);margin-bottom:16px;box-shadow:0 2px 8px rgba(0,0,0,.04)}
.dtable{width:100%;border-collapse:collapse;font-size:12px;background:var(--card)} .dtable thead th{font-fam
ily:var(--mono);font-size:10px;letter-spacing:.1em;text-transform:uppercase;color:var(--text3);background:rg
ba(0,0,0,.02);padding:12px 14px;text-align:right;border-bottom:2px solid
var(--card-border);white-space:nowrap;font-weight:600} .dtable thead th:first-child{text-align:left} .dtable
tbody td{padding:10px 14px;text-align:right;border-bottom:1px solid
rgba(0,0,0,.03);font-family:var(--mono);font-size:12px;color:var(--text)} .dtable tbody

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td:first-child{text-align:left;font-weight:600;color:var(--gas)} .dtable tbody tr:hover
td{background:var(--gas-bg)} .dtable.res tbody td{color:var(--oil)} .dtable.res tbody
td:first-child{color:var(--gas)} .edit-cell{background:#f8fafb;border:1px solid
var(--card-border);border-radius:5px;color:var(--text);font-family:var(--mono);font-size:12px;padding:5px
8px;width:110px;text-align:right;outline:none;transition:all .2s}
.edit-cell:focus{border-color:var(--gas);box-shadow:0 0 2px rgba(10,138,62,.1)}
.sts-select{background:#f8fafb;border:1px solid
var(--card-border);border-radius:5px;color:var(--text);font-family:var(--mono);font-size:10px;padding:5px
8px;outline:none;cursor:pointer;font-weight:700} .sts-select option{background:#fff} /* KPI ROW */
.kpi-row{display:grid;grid-template-columns:repeat(4,1fr);gap:14px;margin:16px 0}
.kpi{background:var(--card);backdrop-filter:blur(16px);border:1.5px solid
var(--card-border);border-radius:16px;padding:20px;position:relative;overflow:hidden;transition:all
.3s;box-shadow:0 4px 16px rgba(0,0,0,.04)}
.kpi:hover{transform:translateY(-6px);border-color:var(--gas-border);box-shadow:0 15px 45px
rgba(10,138,62,.15)} .kpi::before{content:'';position:absolute;top:0;left:0;right:0;height:4px}
.kpi:nth-child(1)::before{background:linear-gradient(90deg,var(--oil),var(--oil-light))}
.kpi:nth-child(2)::before{background:linear-gradient(90deg,var(--gas),var(--gas-light))}
.kpi:nth-child(3)::before{background:linear-gradient(90deg,var(--water),var(--water-light))}
.kpi:nth-child(4)::before{background:linear-gradient(90deg,#e68a00,#ffc107)} .kpi::after{content:'';position
:absolute;bottom:0;right:0;width:60px;height:60px;border-radius:50%;opacity:.04}
.kpi:nth-child(1)::after{background:var(--oil)} .kpi:nth-child(2)::after{background:var(--gas)}
.kpi:nth-child(3)::after{background:var(--water)} .kpi:nth-child(4)::after{background:#ffc107} .kpi-lbl{font
-size:10px;font-weight:600;color:var(--text3);text-transform:uppercase;letter-spacing:.12em;margin-bottom:8p
x} .kpi-val{font-family:var(--mono);font-size:24px;font-weight:800;color:var(--text);letter-spacing:-.02em}
.kpi-u{font-family:var(--mono);font-size:9px;color:var(--text3);letter-spacing:.1em;margin-top:4px} /*
CHARTS */ .chart-box{background:linear-gradient(135deg,#fff,#fafcfe);border:1.5px solid
var(--card-border);border-radius:16px;padding:22px;margin-bottom:18px;box-shadow:0 4px 16px
rgba(0,0,0,.04);position:relative;overflow:hidden} .chart-box::before{content:'';position:absolute;top:0;lef
t:0;right:0;height:3px;background:linear-gradient(90deg,var(--oil),var(--gas),var(--water));opacity:.5} .cha
rt-title{font-family:var(--mono);font-size:11px;letter-spacing:.12em;color:var(--text3);text-transform:upper
case;margin-bottom:14px;font-weight:700;display:flex;align-items:center;gap:8px}

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.chart-title::before{content:'';width:8px;height:8px;border-radius:50%;background:var(--gas);opacity:.6}
.charts-grid{display:grid;grid-template-columns:1fr 1fr;gap:18px} /* CONTROL PANEL */
.control-grid{display:grid;grid-template-columns:repeat(3,1fr);gap:16px;margin-bottom:20px}
.ctrl-card{background:var(--card);border:1px solid
var(--card-border);border-radius:14px;padding:18px;transition:all .3s;box-shadow:0 2px 8px rgba(0,0,0,.04)}
.ctrl-card:hover{border-color:var(--gas-border);box-shadow:0 4px 16px rgba(10,138,62,.08)} .ctrl-head{font-s
ize:13px;font-weight:700;color:var(--text);margin-bottom:10px;display:flex;align-items:center;gap:8px}
.ctrl-row{display:flex;justify-content:space-between;align-items:center;padding:6px
0;font-size:12px;color:var(--text2);border-bottom:1px solid rgba(0,0,0,.03)}
.ctrl-val{font-family:var(--mono);font-size:12px;font-weight:600;color:var(--text)}
.ctrl-bar{height:4px;background:#e8ecf1;border-radius:2px;margin-top:4px;overflow:hidden}
.ctrl-fill{height:100%;border-radius:2px;transition:width .6s ease} /* SUCCESS BANNER */
.success-banner{background:linear-gradient(135deg,var(--gas-bg),rgba(46,204,113,.04));border:1px solid
var(--gas-border);border-radius:12px;padding:16px
20px;font-size:13px;color:var(--gas);font-weight:500;text-align:center;margin-top:20px} /* ===== EQUATIONS
PANEL ===== */ .eq-grid{display:grid;grid-template-columns:1fr 1fr;gap:16px}
.eq-card{background:var(--card);border:1px solid
var(--card-border);border-radius:14px;padding:20px;box-shadow:0 2px 8px rgba(0,0,0,.04)} .eq-card
h3{font-size:14px;color:var(--text);margin-bottom:10px;padding-bottom:8px;border-bottom:2px solid
var(--gas-border)} .eq-formula{background:#f0f4f8;border:1px solid
var(--card-border);border-radius:8px;padding:12px
16px;font-family:var(--mono);font-size:13px;color:var(--oil);margin:8px 0;line-height:1.6;overflow-x:auto}
.eq-desc{font-size:11px;color:var(--text3);margin-bottom:8px;line-height:1.4}
.eq-inputs{display:flex;flex-direction:column;gap:6px;margin-top:10px} .eq-inputs
label{display:grid;grid-template-columns:120px 90px
50px;align-items:center;gap:6px;font-size:11px;color:var(--text2)} .eq-inputs
input{background:#f8fafb;border:1px solid
var(--card-border);border-radius:5px;color:var(--text);font-family:var(--mono);font-size:11px;padding:5px
8px;text-align:right;outline:none} .eq-inputs input:focus{border-color:var(--gas);box-shadow:0 0 2px
rgba(10,138,62,.1)} .eq-result{font-family:var(--mono);font-size:14px;font-weight:700;color:var(--text);marg
in-top:8px;padding:8px 12px;background:var(--gas-bg);border-radius:6px}
.eq-calc-btn{background:var(--gas);color:#fff;border:none;border-radius:6px;padding:8px
16px;font-size:11px;font-weight:700;cursor:pointer;margin-top:8px;transition:all .2s}
.eq-calc-btn:hover{background:var(--gas-light);box-shadow:0 2px 8px rgba(10,138,62,.2)} /* ===== PROCESS
DIAGRAM DETAIL ===== */ .diagram-detail-overlay{display:none;position:fixed;top:0;left:0;right:0;bottom:0;ba
ckground:rgba(0,0,0,.5);z-index:200;align-items:center;justify-content:center;backdrop-filter:blur(4px)}
.diagram-detail-overlay.active{display:flex} .diagram-detail{background:var(--card);border-radius:16px;paddi

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ng:0;max-width:95vw;max-height:90vh;overflow:auto;box-shadow:0 20px 60px rgba(0,0,0,.2);position:relative}
.diagram-detail-header{display:flex;align-items:center;justify-content:space-between;padding:16px
24px;border-bottom:2px solid
var(--gas-border);position:sticky;top:0;background:var(--card);z-index:1;border-radius:16px 16px 0 0}
.diagram-detail-header h2{font-size:16px;color:var(--text);font-weight:700}
.diagram-detail-close{background:none;border:1px solid var(--card-border);color:var(--text2);width:32px;heig
ht:32px;border-radius:8px;cursor:pointer;font-size:18px;display:flex;align-items:center;justify-content:cent
er;transition:all .2s} .diagram-detail-close:hover{background:#fee;color:#c00;border-color:#fcc}
.diagram-detail-body{padding:24px;min-width:900px} /* SVG Popup tooltip */
.svg-popup{display:none;position:absolute;background:var(--card);border:1px solid
var(--gas-border);border-radius:10px;padding:14px 18px;box-shadow:0 8px 30px
rgba(0,0,0,.15);z-index:150;max-width:300px;font-size:12px;line-height:1.5;color:var(--text)}
.svg-popup.show{display:block} .svg-popup h4{color:var(--gas);font-size:13px;margin-bottom:6px} .svg-popup
p{color:var(--text2);margin:0} /* ===== ADDITIVES DONUT CHART ===== */
.additives-chart-wrap{display:flex;align-items:center;gap:24px;margin-top:16px}
.additives-legend{display:flex;flex-direction:column;gap:6px}
.additives-legend-item{display:flex;align-items:center;gap:8px;font-size:11px;color:var(--text2)}
.additives-legend-dot{width:12px;height:12px;border-radius:50%;flex-shrink:0} .additives-legend-val{font-fam
ily:var(--mono);font-weight:600;color:var(--text);margin-left:auto;padding-left:12px} /* VIDEO */
.video-wrap{position:relative;width:100%;padding-bottom:45%;border-radius:12px;overflow:hidden;border:1px
solid var(--card-border);background:#000} .video-wrap
iframe{position:absolute;top:0;left:0;width:100%;height:100%} /* Toast notification for save */
.toast{position:fixed;bottom:30px;right:30px;background:var(--gas);color:#fff;padding:12px 24px;border-radiu
s:10px;font-size:13px;font-weight:600;z-index:999;opacity:0;transform:translateY(20px);transition:all
.3s;pointer-events:none;box-shadow:0 6px 20px rgba(10,138,62,.3)}
.toast.show{opacity:1;transform:translateY(0)}

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/* Touch-friendly */ @media(pointer:coarse){ .nav-btn{padding:14px 16px;font-size:14px;min-height:48px}
.calc-btn{padding:16px;font-size:15px;min-height:52px} .edit-cell,.sts-select{padding:10px
12px;font-size:14px;min-height:44px} .input-group label{font-size:14px;padding:4px 0} .input-group
input{padding:10px 12px;font-size:14px;min-height:44px} } /* LARGE SCREENS */ @media(min-width:1440px){
.content{max-width:1200px} .kpi-row{grid-template-columns:repeat(4,1fr)}
.charts-grid{grid-template-columns:1fr 1fr} } /* TABLET LANDSCAPE */ @media(max-width:1280px){
.sidebar{width:180px} .calc-grid{gap:14px} .kpi-row{grid-template-columns:repeat(2,1fr)}
.eq-grid{grid-template-columns:1fr} } /* TABLET PORTRAIT */ @media(max-width:1024px){
.calc-grid,.charts-grid,.eq-grid{grid-template-columns:1fr} .kpi-row{grid-template-columns:repeat(2,1fr)}
.control-grid{grid-template-columns:1fr 1fr} .sidebar{width:64px;padding:12px 4px}
.nav-btn{font-size:0;padding:12px 8px;text-align:center;justify-content:center;display:flex}
.nav-btn::before{font-size:18px} .content{padding:16px} .header{padding:10px 16px}
.header-title{font-size:16px} .header-sub{font-size:10px} .input-group label{grid-template-columns:1fr 90px
40px;gap:6px} } /* PHONE */ @media(max-width:768px){ .layout{flex-direction:column} .sidebar{
width:100%;flex-direction:row;overflow-x:auto;height:auto; position:sticky;top:50px;z-index:99; padding:6px
8px;gap:4px; background:#fff;border-bottom:1px solid var(--card-border); -webkit-overflow-scrolling:touch; }
.nav-btn{font-size:11px;white-space:nowrap;padding:10px 14px;flex-shrink:0}
.content{padding:12px;max-height:none;overflow-y:visible} .kpi-row{grid-template-columns:1fr}
.control-grid{grid-template-columns:1fr} .calc-grid,.eq-grid{grid-template-columns:1fr}
.charts-grid{grid-template-columns:1fr}
.header{flex-direction:column;gap:8px;align-items:flex-start;padding:10px 14px}
.header-title{font-size:15px} .header-right{width:100%;justify-content:flex-end}
.panel-title{font-size:15px}

```

```

.dtable{font-size:11px} .dtable thead th{padding:8px 6px;font-size:9px} .dtable tbody td{padding:8px 6px}
.edit-cell{width:80px;font-size:11px} .input-group label{grid-template-columns:1fr;gap:4px} .input-group
input{width:100%} .input-group span{display:inline} .res-row{flex-wrap:wrap;gap:2px}
.kpi-val{font-size:18px} .diagram-detail-body{min-width:auto} .additives-chart-wrap{flex-direction:column} }
/* Very small phones */ @media(max-width:480px){ .header-title{font-size:13px} .header-sub{display:none}
.nav-btn{font-size:10px;padding:8px 10px} .panel-title{font-size:14px} .kpi{padding:12px}
.kpi-val{font-size:16px} .edit-cell{width:70px;font-size:10px} } /* ===== SIMULATION & ALARMS & PID ===== */
.sim-btn{background:rgba(0,0,0,0.3);border:1px solid
rgba(0,255,136,0.5);color:#00e676;font-size:11px;font-weight:700;padding:6px
14px;border-radius:6px;cursor:pointer;font-family:'Inter',sans-serif;transition:all
.2s;letter-spacing:.03em;display:flex;align-items:center;gap:6px;}
.sim-btn.active{background:rgba(0,255,136,0.15);color:#fff;border-color:#00ff88;animation:glow 2s infinite;}
.pid-wrap{background:linear-gradient(135deg,var(--card),#fafcfe);border:1.5px solid
var(--card-border);border-radius:16px;padding:22px;margin-bottom:20px;box-shadow:0 4px 16px
rgba(0,0,0,.04);} .pid-flex{display:flex;gap:30px;align-items:center;} .pid-tank-wrap{flex:0 0
100px;display:flex;justify-content:center;}
.pid-controls{flex:1;display:flex;flex-direction:column;gap:8px;} .pid-controls
input[type=range]{width:100%;accent-color:var(--gas);margin-bottom:4px;cursor:pointer;}

```

```
.alarm-dropdown-container { position: relative; display: flex; align-items: center; } .badge-mini {
position: absolute; top: -4px; right: -4px; background: #ff5252; color: #fff; font-size: 10px; padding: 2px
5px; border-radius: 10px; font-family: var(--mono); font-weight: bold; border: 1px solid #fff; box-shadow: 0
0 5px rgba(255,82,82,0.5); } .alarm-panel { position: absolute; top: 45px; right: 0; width: 320px;
background: var(--card);
```

```
border: 1.5px solid #ff5252; border-radius: 12px; box-shadow: 0 10px 40px rgba(0,0,0,0.2); z-index: 1000;
transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1); display: flex; flex-direction: column; max-height: 450px;
overflow: hidden; opacity: 1; transform: translateY(0); } .alarm-panel.minimized { opacity: 0; transform:
translateY(-10px); pointer-events: none; } .alarm-head { background:
linear-gradient(135deg,#ff5252,#c62828); color: #fff; padding: 12px 16px; font-weight: 700; font-size: 14px;
display: flex; justify-content: space-between; align-items: center; cursor: pointer; } .badge { background:
#fff; color: #c62828; padding: 2px 8px; border-radius: 12px; font-size: 11px; font-family: var(--mono); }
.alarm-list{padding:10px;overflow-y:auto;display:flex;flex-direction:column;gap:8px;}
.alarm-item{background:#fff3f3;border-left:4px solid #ff5252;padding:10px;font-size:12px;color:#c62828;bord
er-radius:4px;display:flex;flex-direction:column;gap:4px;animation:fadeIn 0.3s ease;}
.alarm-item.warning{background:#fff8e1;border-color:#ffab40;color:#ef6c00;}
.alarm-time{font-family:var(--mono);font-size:10px;opacity:0.8;} html[dir="rtl"]
.sim-btn{flex-direction:row-reverse;} html[dir="rtl"] .pid-flex{flex-direction:row-reverse;} html[dir="rtl"]
.alarm-panel{right:auto;left:0;} html[dir="rtl"] .alarm-item{border-left:none;border-right:4px solid
#ff5252;} html[dir="rtl"] .alarm-item.warning{border-right-color:#ffab40;} /* BLUEPRINT SCHEMATIC */
.blueprint-wrap { width: 100%; overflow-x: auto; padding: 40px 0; display: flex; justify-content: center;
background: radial-gradient(circle at center, #0a1b3d, #061021); border-radius: 12px; } .blueprint-wrap svg
{ width: 90%; height: auto; min-width: 800px; filter: drop-shadow(0 0 10px rgba(0,240,255,0.1)); }
.blueprint-wrap rect, .blueprint-wrap circle, .blueprint-wrap ellipse { transition: all 0.3s; cursor:
pointer; } .blueprint-wrap g:hover rect { filter: brightness(1.2); stroke-width: 2.5; } .blueprint-wrap text
{ font-family: var(--mono); pointer-events: none; } /* Print */ @media print{
.sidebar,.header-right,.calc-btn,.nav-btn,.panel-nav,.alarm-panel{display:none}
body{background:#fff;color:#000} .layout{display:block} .content{max-height:none;overflow:visible}
```

```
.tab-panel{display:block!important;page-break-after:always} }
```

■ File: login.css

```
/* ===== LOGIN SCREEN ===== */ @keyframes
loginFloat{0%,100%{transform:translateY(0)}50%{transform:translateY(-12px)}} @keyframes
loginPulse{0%,100%{opacity:.6;transform:scale(1)}50%{opacity:1;transform:scale(1.5)}} @keyframes
loginGlow{0%{box-shadow:0 0 30px rgba(0,240,255,.15),0 0 60px rgba(0,230,118,.08)}50%{box-shadow:0 0 50px
rgba(0,240,255,.3),0 0 90px rgba(124,77,255,.15)}100%{box-shadow:0 0 30px rgba(0,240,255,.15),0 0 60px
rgba(0,230,118,.08)}} @keyframes loginShake{0%,100%{transform:translateX(0)}15%{transform:translateX(-12px)}
30%{transform:translateX(10px)}45%{transform:translateX(-8px)}60%{transform:translateX(6px)}75%{transform:tr
anslateX(-3px)}} @keyframes
loginSuccess{0%{border-color:rgba(0,230,118,.3)}50%{border-color:#00e676;box-shadow:0 0 60px
rgba(0,230,118,.4)}100%{border-color:rgba(0,230,118,.3)}} @keyframes
ringRotate{from{transform:rotate(0deg)}to{transform:rotate(360deg)}} @keyframes
ringRotateR{from{transform:rotate(360deg)}to{transform:rotate(0deg)}} @keyframes colorShift{0%{border-color:
#00f0ff;filter:hue-rotate(0deg)}25%{border-color:#00e676;filter:hue-rotate(60deg)}50%{border-color:#7c4dff;f
ilter:hue-rotate(120deg)}75%{border-color:#ff6d00;filter:hue-rotate(180deg)}100%{border-color:#00f0ff;filter
:hue-rotate(360deg)}} @keyframes
dotPop{0%{transform:scale(0);opacity:0}50%{transform:scale(1.3)}100%{transform:scale(1);opacity:1}}
@keyframes fadeOutLogin{0%{opacity:1;transform:scale(1)}100%{opacity:0;transform:scale(1.05)}}
#loginScreen{position:fixed;top:0;left:0;right:0;bottom:0;z-index:10000}
#loginScreen.fadeout{animation:fadeOutLogin .6s ease forwards}
.login-bg{width:100%;height:100%;display:flex;align-items:center;justify-content:center;
background:linear-gradient(135deg,#060912 0%,#0a1428 50%,#060912 100%); position:relative;overflow:hidden}
.login-particles{position:absolute;inset:0;pointer-events:none}
.login-particle{position:absolute;border-radius:50%;opacity:0;animation:loginPulse var(--dur,4s) ease-in-out
infinite;animation-delay:var(--delay,0s)} .login-rings{position:absolute;inset:0;display:flex;align-items:ce
nter;justify-content:center;pointer-events:none} .login-ring{position:absolute;border-radius:50%;border:1px
solid transparent} .ring1{width:600px;height:600px;border-color:rgba(0,240,255,.05);animation:ringRotate 30s
linear infinite}
.ring2{width:450px;height:450px;border-color:rgba(0,230,118,.07);border-style:dashed;animation:ringRotateR
25s linear infinite} /* Card */
.login-card{position:relative;z-index:10;background:rgba(10,15,28,.8);backdrop-filter:blur(40px); border:1px
solid rgba(0,240,255,.2);border-radius:32px;display:flex;width:880px;max-width:95vw;min-height:500px;
box-shadow:0 25px 80px rgba(0,0,0,0.5); animation:loginGlow 5s ease-in-out infinite;overflow:hidden;
transition:all .5s} .login-card.shake{animation:loginShake .5s ease}
.login-card.success{animation:loginSuccess .8s ease}
.login-left{flex:1.2;padding:50px;display:flex;flex-direction:column;justify-content:center;
background:linear-gradient(180deg, rgba(0,240,255,0.03) 0%, transparent 100%); border-right:1px solid
rgba(255,255,255,0.05);position:relative} .login-right{flex:1;padding:50px;display:flex;flex-direction:colum
n;justify-content:center;background:rgba(0,0,0,0.2)} /* Credits Section */
.login-credits-wrap{text-align:center;font-family:'Inter',sans-serif} .university-logo{width:80px;height:80p
x;margin-bottom:24px;background:#fff;border-radius:14px;padding:8px;box-shadow:0 0 25px
rgba(0,240,255,0.3);object-fit:contain} .credits-title{color:#00f0ff;font-size:14px;font-weight:700;letter-s
pacing:.1em;text-transform:uppercase;margin-bottom:15px;opacity:0.8}
.credits-text{color:#e0f0ff;font-size:16px;line-height:1.8;margin-bottom:20px;font-weight:400} .credits-text
strong{display:block;margin-top:15px;color:#00e676;font-size:18px;text-shadow:0 0 10px rgba(0,230,118,0.3)}
/* Logo Section */
.login-logo-wrap{display:flex;flex-direction:column;align-items:center;margin-bottom:30px} .main-logo{width:
100px;height:100px;object-fit:contain;margin-bottom:15px;background:#fff;border-radius:18px;padding:10px;box
-shadow:0 0 30px rgba(0,240,255,0.3)} .station-logo-crop{height:80px;width:auto;object-fit:cover;object-posi
tion:top;background:#fff;padding:8px;border-radius:14px;}
.login-title{font-family:'Inter',sans-serif;font-size:32px;font-weight:900;text-align:center;
background:linear-gradient(135deg,#00f0ff,#00e676);
-webkit-background-clip:text;-webkit-text-fill-color:transparent;background-clip:text; margin-bottom:5px} .l
ogin-subtitle{text-align:center;font-size:11px;color:#5a7aaa;letter-spacing:.3em;text-transform:uppercase;fo
nt-weight:600} /* Form */

.login-field{margin-bottom:20px} .login-label{display:block;font-size:10px;color:#5a7aaa;text-transform:uppe
rcase;letter-spacing:.2em;font-weight:700;margin-bottom:10px}
.login-input-wrap{display:flex;align-items:center;background:rgba(0,0,0,0.4);border:1px solid
rgba(0,240,255,0.2);border-radius:14px;transition:all .3s}
.login-input-wrap:focus-within{border-color:#00f0ff;box-shadow:0 0 20px
rgba(0,240,255,0.15);background:rgba(0,0,0,0.6)}
.login-lock-icon{padding-left:15px;font-size:18px;opacity:.7}
.login-input{flex:1;background:none;border:none;outline:none;color:#fff;font-family:'JetBrains
Mono',monospace;font-size:18px;padding:15px;letter-spacing:.2em}
.login-input::placeholder{color:rgba(255,255,255,0.15);letter-spacing:.1em}
.login-btn{width:100%;margin-top:10px;padding:16px;background:linear-gradient(135deg,#00f0ff,#00e676); borde
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r:none;border-radius:14px;cursor:pointer;display:flex;align-items:center;justify-content:center;gap:12px;transition:all .3s} .login-btn:hover{transform:translateY(-3px);box-shadow:0 10px 30px
rgba(0,240,255,0.3);filter:brightness(1.1)}
.login-btn-text{color:#060912;font-weight:800;letter-spacing:0.1em;font-size:14px}
.login-btn-arrow{color:#060912;font-size:20px} .login-error{text-align:center;font-size:12px;font-weight:600
;margin-top:15px;height:20px;color:#ff5252;opacity:0} .login-error.show{opacity:1}
.login-error.ok{color:#00e676} /* Responsive */ @media(max-width:900px){
.login-card{flex-direction:column;width:95vw;min-height:auto}
.login-left{padding:30px;border-right:none;border-bottom:1px solid rgba(255,255,255,0.05)}
.login-right{padding:30px} .login-title{font-size:24px} }
```

■ File: core.js

```
"use strict"; // ===== i18n SYSTEM ===== var currentLang='en'; var i18n={ en:{overview_title:'Plant Overview
- Process Flow Schematic',svg_wellhead:'WELLHEAD',svg_manifold:'MANIFOLD',svg_hpsep:'HP
SEPARATOR',svg_lpsep:'LP SEPARATOR',svg_gas:'GAS TREATMENT',svg_gasexp:'GAS EXPORT',svg_oil:'OIL
TREATMENT',svg_water:'WATER TREATMENT',svg_storage:'STORAGE',svg_export:'EXPORT',svg_chem:'CHEMICAL
INJ.',svg_leg_oil:'Oil Flow',svg_leg_gas:'Gas Flow',svg_leg_water:'Water Flow',svg_leg_chem:'Chemical Inj.',
nav_overview:'■ Overview',nav_separator:'■■ Separator Calc',nav_trains:'■ Production Trains',nav_gas:'■ Gas
Treatment',nav_oil:'■■ Oil Treatment',nav_water:'■ Water Treatment',nav_chemical:'■ Chemical
Injection',nav_storage:'■■ Storage & Export',nav_charts:'■ Charts',nav_equations:'■
Equations',nav_diagram:'■ Process Diagram',nav_control:'■ Control Panel',
btn_lang:'■■■■■■■■■■',success_msg:'■ All systems are stable. Mass balance and chemical injection rates are
automatically updated based on live input.', status_online:'ONLINE',alarms_title:'System
Alarms',pid_title:'PID Level Control - HP Separator',pid_valve:'Liquid Outflow Valve
(%)',pid_retention:'Retention Time (min)',pid_auto:'Auto Target 50%', sim_on:'■■ Live: ON',sim_off:'■■
Live: OFF'}, ar:{overview_title:'■■■■ ■■■■ ■■■■ ■■■■■■ - ■■■■ ■■■■■■■■■■',svg_wellhead:'■■■■
■■■■■',svg_manifold:'■■■■■■■■■■■■■■■■■■■■',svg_hpsep:'■■■■■ ■. ■■■■■',svg_lpsep:'■■■■■ ■. ■■■■■',svg_gas:'■■■■■■■■
■■■■■■',svg_gasexp:'■■■■■■ ■■■■■',svg_oil:'■■■■■■ ■■■■■',svg_water:'■■■■■■■■
■■■■■■',svg_storage:'■■■■■■■■■■',svg_export:'■■■■■■■■■■',svg_chem:'■■■■ ■■■■■■■■■',svg_leg_oil:'■■■■■
■■■■■■',svg_leg_gas:'■■■■■ ■■■■■',svg_leg_water:'■■■■■ ■■■■■',svg_leg_chem:'■■■■ ■■■■■■■■■',
nav_overview:'■ ■■■■ ■■■■',nav_separator:'■■ ■■■■■■ ■■■■■',nav_trains:'■ ■■■■■■■■
■■■■■■■■',nav_gas:'■ ■■■■■■ ■■■■■',nav_oil:'■■ ■■■■■■ ■■■■■',nav_water:'■ ■■■■■■■■
■■■■■■',nav_chemical:'■ ■■■■■ ■■■■■■■■■■■',nav_storage:'■ ■■■■■ ■■■■■■■■■■■',nav_charts:'■ ■■■■■■■■
■■■■■■■■',nav_equations:'■ ■■■■■■■■■■■',nav_diagram:'■ ■■■■ ■■■■■■■■■',nav_control:'■ ■■■■ ■■■■■■■■■',
btn_lang:'English',success_msg:'■ ■■■■ ■■■■■■■■■ ■■■■■. ■■■ ■■■■■ ■■■■■ ■■■■■ ■■■■■ ■■■■■
■■■■■■■■ ■■■■■■■ ■■■■■ ■■■ ■■■■■■■■■.', status_online:'■■■■ ■■■■',alarms_title:'■■■■■■■■■
■■■■■■■■',pid_title:'■■■■■■■■ ■■■■■■■■■ PID - ■■■■■ ■■■■■ ■■■■■',pid_valve:'■■■■ ■■■■ ■■■■■
(%)',pid_retention:'■■■ ■■■■■■ (■■■■■■)',pid_auto:'■■■ ■■■■■■ 50%', sim_on:'■■ ■■ ■■■■■:
■■■■',sim_off:'■■ ■■ ■■■■■: ■■■■■'} }; function applyLang(){ var t=i18n[currentLang];
document.querySelectorAll('[data-i18n]').forEach(function(el){ var key=el.getAttribute('data-i18n');
if(t[key])el.textContent=t[key]; }); // Sidebar buttons var tabs=['overview','separator','trains','gas','oil
','water','chemical','storage','charts','equations','diagram','control'];
document.querySelectorAll('.nav-btn').forEach(function(b,i){ var key='nav_'+tabs[i];
if(t[key])b.textContent=t[key]; }); // Lang button $('langBtn').textContent=t.btn_lang; // Success banner
document.querySelector('.success-banner').forEach(function(el){el.textContent=t.success_msg}); // RTL if(
currentLang==='ar'){document.documentElement.setAttribute('dir','rtl');document.documentElement.setAttribute
('lang','ar')}}
else{document.documentElement.setAttribute('dir','ltr');document.documentElement.setAttribute('lang','en')}
} function toggleLang(){ currentLang=currentLang==='en'?'ar':'en'; applyLang(); } var trainData=[ {id:"TRAIN
1",status:"ON",inlet:31731.52,oil:22513.10,water:9218.42,gas:64.8,sepVol:60}, {id:"TRAIN
2",status:"ON",inlet:46508.43,oil:37796.91,water:8711.52,gas:36.1,sepVol:60}, {id:"TRAIN
3",status:"ON",inlet:51067.11,oil:39528.68,water:11538.43,gas:35.9,sepVol:60}, {id:"TRAIN
4",status:"STOP",inlet:0,oil:0,water:0,gas:0,sepVol:60} ]; function fmt(n){return
n.toLocaleString("en-US",{minimumFractionDigits:2,maximumFractionDigits:2})} function $(id){return
document.getElementById(id)} function val(id){return parseFloat($(id).value)||0} // Clock
setInterval(function(){var d=new Date();$("#clock").textContent=d.toLocaleString("en-GB",{day:"2-digit",month
:"short",year:"numeric",hour:"2-digit",minute:"2-digit",second:"2-digit"}),1000); // Tabs function
switchTab(t){ document.querySelectorAll(".tab-panel").forEach(function(p){p.classList.remove("active")});
document.querySelectorAll(".nav-btn").forEach(function(b){b.classList.remove("active")}); var
panel=$( "tab-"+t);if(panel)panel.classList.add("active");
document.querySelectorAll('.nav-btn[data-tab="'+t+'"]').forEach(function(b){b.classList.add("active")}); }
document.querySelectorAll(".nav-btn").forEach(function(b){
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b.addEventListener("click",function(){switchTab(this.dataset.tab)}); }); // Result row helper function
rr(label,value,unit){return '<div class="res-row"><span class="res-label">'+label+'</span><span><span
class="res-value">'+value+'</span><span class="res-unit"> '+unit+'</span></span></div>'} // KPI helper
function kpi(lbl,val,unit){return '<div class="kpi"><div class="kpi-lbl">'+lbl+'</div><div
class="kpi-val">'+val+'</div><div class="kpi-u">'+unit+'</div></div>'} // ===== TRAIN TABLE ===== function
renderTrains(){ var tb=$( "trainBody");tb.innerHTML=""; trainData.forEach(function(t,i){ var
tr=document.createElement("tr"); var h='<td>'+t.id+'</td>'; h+='<td><select class="sts-select"
onchange="trainSts(''+i+'',this.value)"><option value="ON"' + (t.status=="ON"?
selected:"")>ON</option><option value="STOP"' + (t.status=="STOP"?
selected:"")>STOP</option></select></td>'; ["inlet","oil","water","gas","sepVol"].forEach(function(f){
h+='<td><input class="edit-cell" type="number" value="'+t[f]+'"' step="0.01" min="0"
onchange="trainEdit(''+i+'',''+f+'',this.value)"></td>'; }); tr.innerHTML=h;tb.appendChild(tr); });
renderTrainResults(); } function trainSts(i,v){trainData[i].status=v;if(v=="STOP"){trainData[i].inlet=0;tra
inData[i].oil=0;trainData[i].water=0;trainData[i].gas=0}renderTrains()} function
trainEdit(i,f,v){trainData[i][f]=parseFloat(v)||0;renderTrainResults()} function renderTrainResults(){ var
```

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tb=$( "trainResults" ); tb.innerHTML=""; var tI=0, tO=0, tG=0; trainData.forEach(function(d){ var
wc=d.inlet>0?(d.water/d.inlet)*100:0; var gor=d.oil>0?(d.gas*1e6)/d.oil:0; var im=d.inlet*0.15898; var
rt=im>0?(d.sepVol*1440)/im:0; tb.innerHTML+=<tr><td>' + d.id + '</td><td>' + fmt(wc) + '</td><td>' + fmt(gor) + '</td><
td>' + fmt(im) + '</td><td>' + fmt(rt) + '</td><tr>'; tI+=d.inlet; tO+=d.oil; tG+=d.gas; }); var
dem=((tI*158.98)*(15/1e6))/24; var eng=tG*1000; $( "trainKPI" ).innerHTML=kpi("Total
Inlet", fmt(tI), "bpd") + kpi("Total Oil", fmt(tO), "bpd") + kpi("Demulsifier Rate", fmt(dem), "L/hr") + kpi("Gas
Energy", fmt(eng), "MMBTU/D"); updateOverview(tI, tO, tG); } // ===== OVERVIEW ===== function
updateOverview(tI, tO, tG){ var tw=trainData.reduce(function(s,d){return s+d.water}, 0); var
dem=((tI*158.98)*(15/1e6))/24; // Update SVG schematic values var ids={well:fmt(tI)+'
bpd', manifold:fmt(tI)+' bpd', hpsep:fmt(tI)+' bpd', lpsep:fmt(tO)+' bpd', gas:fmt(tG)+' MMSCFD', oil:fmt(tO)+'
bpd', water:fmt(tw)+' bpd', storage:fmt(tO*5.615)+' bbl', export:fmt(tO)+' bpd'}; for(var k in ids){var
el=$( 'sv-' + k ); if(el) el.textContent=ids[k]} $( "overviewKPI" ).innerHTML=kpi("Total
Inlet", fmt(tI), "bpd") + kpi("Total Oil", fmt(tO), "bpd") + kpi("Total
Gas", fmt(tG), "MMSCFD") + kpi("Demulsifier", fmt(dem), "L/hr"); // Update new summary panel var sumInlet =
$( 'sum-inlet' ); if(sumInlet) sumInlet.innerHTML= fmt(tI) + ' <span style="font-size:10px;">bpd</span>'; }
// ===== SEPARATOR CALCULATOR (API 12J) ===== function calcSeparator(){ var
P=val("sep-P"), T=val("sep-T"), Qo=val("sep-Qo"), Qw=val("sep-Qw"), Qg=val("sep-Qg"); var
API=val("sep-API"), SGg=val("sep-SGg"), mu0=val("sep-mu0"), dp=val("sep-dp"), K=val("sep-K");

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var tro=val("sep-tro"), trw=val("sep-trw"); // Oil density from API var SGo=141.5/(API+131.5); var
rho0=SGo*62.4; // lb/ft3 var rhoG=SGg*0.0764*(P+14.7)/14.7*(520/(T+460)); // real gas approx var rhoW=64.3;
// lb/ft3 water // Souders-Brown gas velocity var Vmax=K*Math.sqrt((rho0-rhoG)/rhoG); // Gas volumetric at
conditions var Qg_acf=Qg*1e6/86400*(14.7/(P+14.7))*((T+460)/520); // actual ft3/s var
Ag=Qg_acf>0?Qg_acf/Vmax:0; // gas area ft2 var Dgas=Math.sqrt(4*Ag/Math.PI); // Liquid retention sizing var
QL_ft3min=(Qo+Qw)*5.615/1440; var Vliq=QL_ft3min*(tro+trw); // ft3 liquid holdup // Assume 50% liquid level,
horizontal var Dliq=Math.pow(Vliq*8/(Math.PI*3), 1/3); // D with L=3D var D=Math.max(Dgas, Dliq); var L=3*D;
var Across=Math.PI/4*D*D; var Vtotal=Across*L; var Vliq_actual=0.5*Vtotal; // Stokes settling velocity var
dpM=dp*1e-6*3.281; // micron to ft var Vt=((rhoW-rho0)*32.174*dpM*dpM)/(18*mu0*6.72e-4); // ft/s // Water
cut var WC=(Qo+Qw)>0?Qw/(Qo+Qw)*100:0; var GOR=Qo>0?(Qg*1e6)/Qo:0; // Retention actual var
retActual=QL_ft3min>0?Vliq_actual/QL_ft3min:0; var r=$( "sepResults" ); r.innerHTML=rr("Oil Density
(po)", fmt(rho0), "lb/ft3") + rr("Gas Density (pg)", fmt(rhoG), "lb/ft3") + rr("Gas Velocity
(Vmax)", fmt(Vmax), "ft/s") + rr("Stokes Settling", fmt(Vt*12), "in/s") + rr("Water
Cut", fmt(WC), "%") + rr("GOR", fmt(GOR), "SCF/bbl") + rr("Retention Time", fmt(retActual), "min"); var
v=$( "sepVessel" ); v.innerHTML=rr("Vessel Diameter", fmt(D)+" ft / "+fmt(D*12), "in") + rr("Seam-to-Seam
Length", fmt(L), "ft") + rr("Cross Section Area", fmt(Across), "ft2") + rr("Total
Volume", fmt(Vtotal), "ft3") + rr("Liquid Volume (50%)", fmt(Vliq_actual), "ft3") + rr("Liquid
Volume", fmt(Vliq_actual*0.178107), "bbl"); buildSepChart(P, D, L, Vtotal); } // ===== GAS TREATMENT =====
function calcGas(){ var Q=val("gas-Q"), Pin=val("gas-Pin"), Tin=val("gas-Tin"), SG=val("gas-SG"); var
H2S=val("gas-H2S"), CO2=val("gas-CO2"), H2O=val("gas-H2O"), Pout=val("gas-Pout"); // Gas MW var MW=SG*28.97; //
Z factor approx (Hall-Yarborough simplified) var Tpc=168+325*SG-12.5*SG*SG; var Ppc=677+15*SG-37.5*SG*SG;
var Tpr=(Tin+460)/Tpc; var Ppr=(Pin+14.7)/Ppc; var
Z=1-(3.53*Ppr)/(Math.pow(10, 0.9813*Tpr))+0.274*Ppr*Ppr/(Math.pow(10, 0.8157*Tpr));
Z=Math.max(0.3, Math.min(1.2, Z)); // Gas density var rhoG=(Pin+14.7)*MW/(Z*10.73*(Tin+460)); // Compression
ratio var CR=Pout>0?(Pout+14.7)/(Pin+14.7):1; var nStages=Math.max(1, Math.ceil(Math.log(CR)/Math.log(3.5)));
var rPerStage=Math.pow(CR, 1/nStages); // Compression HP (approx) var
HP=Q*1e6/86400*144*(Pin+14.7)/(33000)*nStages*(Math.pow(rPerStage, 0.286)-1)/0.286*1.4; HP=Math.abs(HP); //
Dehydration - glycol rate (2-3 gal TEG/lb H2O)

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var H2Olb=H2O*Q; // lb/day water removed var glycol=H2Olb*3; // gal TEG/day // Amine for H2S/CO2 (approx 3
gal/min per MMSCFD acid gas) var acidGas=(H2S+CO2)/100*Q; var amineRate=acidGas*3*1440; // gal/day //
Thermal energy var energy=Q*1000; // MMBTU/D (HHV-1000 BTU/SCF) var r=$( "gasResults" ); r.innerHTML=rr("Gas
MW", fmt(MW), "g/mol") + rr("Z Factor", Z.toFixed(4), "") + rr("Gas Density", fmt(rhoG), "lb/ft3") + rr("Compression
Ratio", fmt(CR), ":1") + rr("Stages
Required", nStages, "stages") + rr("Ratio/Stage", fmt(rPerStage), ":1") + rr("Compression
Power", fmt(HP), "HP") + rr("Water Removed", fmt(H2Olb), "lb/day") + rr("TEG
Circulation", fmt(glycol), "gal/day") + rr("Amine Rate", fmt(amineRate), "gal/day") + rr("Acid Gas
Load", fmt(acidGas), "MMSCFD") + rr("Thermal Energy", fmt(energy), "MMBTU/D");
buildGasChart(Q, Pin, nStages, energy); } // ===== OIL TREATMENT ===== function calcOil(){ var
Q=val("oil-Q"), API=val("oil-API"), BSWin=val("oil-BSWin"), BSWout=val("oil-BSWout"); var salt=val("oil-salt"),
saltOut=val("oil-saltOut"), Tin=val("oil-Tin"), Tout=val("oil-Tout"), wash=val("oil-wash"); var
SGo=141.5/(API+131.5); var rho0=SGo*62.4; // Water removed var waterRemoved=Q*(BSWin-BSWout)/100; // Net oil
var netOil=Q*(1-BSWout/100); // Heater duty Q=m*Cp*dT, Cp=0.5 BTU/lb/F for crude var
massRate=Q*5.615*rho0/24; // lb/hr var heaterDuty=massRate*0.5*(Tout-Tin)/1e6; // MMBTU/hr // Wash water
volume var washVol=Q*wash/100; // Salt removal efficiency var saltEff=salt>0?(1-saltOut/salt)*100:0; //
Desalter stages (single or two stage) var stages=saltOut<10&&salt>30?2:1; // Electrostatic field (typical
1-2 kV/cm, power ~0.5 kW/1000 bbl) var desPower=Q/1000*0.5*stages; var r=$( "oilResults" );
r.innerHTML=rr("Oil SG", SGo.toFixed(4), "") + rr("Oil Density", fmt(rho0), "lb/ft3") + rr("Water
Removed", fmt(waterRemoved), "bpd") + rr("Net Oil Export", fmt(netOil), "bpd") + rr("Heater
Duty", fmt(heaterDuty), "MMBTU/hr") + rr("Heater Duty (daily)", fmt(heaterDuty*24), "MMBTU/D") + rr("Wash

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Water",fmt(washVol),"bpd")+rr("Salt Removal Eff.",fmt(saltEff),"%")+rr("Desalter
Stages",stages,"stage(s)")+rr("Desalter Power",fmt(desPower),"kW")+rr("Mass Flow
Rate",fmt(massRate),"lb/hr"); buildOilChart(Q,BSwin,BSWout,heaterDuty); } // ===== WATER TREATMENT =====
function calcWater(){ var Q=val("wat-Q"),OiW=val("wat-OiW"),OiWout=val("wat-OiWout"),TSS=val("wat-TSS"),T=va
l("wat-T"),ret=val("wat-ret"); // Skim tank volume var Qft3=Q*5.615/1440; // ft3/min var Vskim=Qft3*ret; var
Dskim=Math.pow(Vskim*4/(Math.PI*2),1/3); // Oil removed var oilRemoved=Q*5.615*(OiW-OiWout)/1e6*7.48; //
gal/day approx // Removal efficiency var remEff=OiW>0?(1-OiWout/OiW)*100:0; // Stokes rise rate for oil in
water var dpOil=60; // micron typical var rho0=52; var rhoW=64.3; var muW=0.7; // cp at ~120F var
dpFt=dpOil*1e-6*3.281; var Vrise=((rhoW-rho0)*32.174*dpFt*dpFt)/(18*muW*6.72e-4); // Hydrocyclone sizing
(typ 500-2000 bpd each) var nHC=Math.max(1,Math.ceil(Q/1500)); var r=$("#waterResults"); r.innerHTML+=rr("Skim
Tank Volume",fmt(Vskim),"ft³")+rr("Skim Tank Volume",fmt(Vskim*0.178107),"bbl")+rr("Skim Tank
Diameter",fmt(Dskim),"ft")+rr("Oil Removal Eff.",fmt(remEff),"%")+rr("Oil
Recovered",fmt(oilRemoved),"gal/day")+rr("Stokes Rise Rate",fmt(Vrise*12),"in/s")+rr("Hydrocyclones
Req.",nHC,"units")+rr("TSS Inlet",fmt(TSS),"mg/L")+rr("Est. TSS Outlet",fmt(TSS*0.15),"mg/L");
buildWaterChart(Q,OiW,OiWout,remEff); }

```

```

// ===== CHEMICAL INJECTION ===== function calcChemical(){ var Ql=val("chem-Ql"); var doses={Demulsifier:val
("chem-demul"),Corrosion_Inh:val("chem-corr"),Scale_Inh:val("chem-scale"),H2S_Scavenger:val("chem-h2s"),Bioc
ide:val("chem-bio"),O2_Scavenger:val("chem-o2"),PPD:val("chem-ppd")}; var r=$("#chemResults"); r.innerHTML="";
var labels=[],vals=[],colors=["#00f0ff","#00e676","#ff9800","#e91e63","#9c27b0","#00bcd4","#ffc107"]; var
total=0,i=0; for(var k in doses){ var d=doses[k]; // rate =
Ql(bpd)*158.98(L/bbl)*dose(ppm=mg/L)/1e6(g/mg)*1000(mL/L) => L/day var Lday=Ql*158.98*d/1e6; var
Lhr=Lday/24; var galDay=Lday*0.264172; r.innerHTML+=rr(k.replace(/_/g," "),fmt(Lhr)+" L/hr |
"+fmt(galDay),"gal/day"); labels.push(k.replace(/_/g," "));vals.push(Lhr;total+=Lhr;i++; }
r.innerHTML+=rr("TOTAL Injection Rate",fmt(total),"L/hr"); r.innerHTML+=rr("TOTAL
Daily",fmt(total*24),"L/day"); buildChemChart(labels,vals,colors); } // ===== STORAGE & EXPORT =====
function calcStorage(){ var
n=val("stg-n"),D=val("stg-D"),H=val("stg-H"),fill=val("stg-fill"),exp=val("stg-exp"),prod=val("stg-prod");
var Vtank=Math.PI*4*D*D*H; // m3 per tank var Vuse=Vtank*fill/100; var Vtotal=Vuse*n; var
Vbbl=Vtotal*6.28981; // m3 to bbl var daysStorage=exp>0?Vbbl/exp:0; var balance=prod-exp; var
r=$("#storageResults"); r.innerHTML+=rr("Tank Volume (each)",fmt(Vtank),"m³")+rr("Usable Volume
(each)",fmt(Vuse),"m³")+rr("Total Usable Volume",fmt(Vtotal),"m³")+rr("Total Usable
Volume",fmt(Vbbl),"bbl")+rr("Storage Days",fmt(daysStorage),"days")+rr("Production
Rate",fmt(prod),"bpd")+rr("Export Rate",fmt(exp),"bpd")+rr("Daily Balance",fmt(balance),"bpd")+rr("Tank
Diameter",fmt(D),"m")+rr("Tank Height",fmt(H),"m"); } // ===== CONTROL PANEL ===== function renderControl(){
var g=$("#controlGrid"); var sections=[ {icon:"■■■",name:"Wellhead",items:[["Well Pressure","2500
psig","85"],["Well Temperature","180 °F","72"],["Choke Size","48/64 in","75"]]}, {icon:"■■■",name:"HP
Separator",items:[["Operating Pressure","150 psig","90"],["Temperature","120 °F","80"],["Liquid
Level","55%","55"]]}, {icon:"■■■",name:"LP Separator",items:[["Operating Pressure","50
psig","88"],["Temperature","110 °F","75"],["Liquid Level","50%","50"]]}, {icon:"■",name:"Gas
Compression",items:[["Suction Pressure","50 psig","70"],["Discharge Pressure","1000 psig","82"],["Power
Load","2500 HP","78"]]}, {icon:"■■■",name:"Oil Heater",items:[["Inlet Temp","100 °F","65"],["Outlet
Temp","150 °F","80"],["Duty","25 MMBTU/hr","72"]]}, {icon:"■",name:"Water Treatment",items:[["OiW
Inlet","500 ppm","60"],["OiW Outlet","35 ppm","92"],["Skim Level","45%","45"]]}, {icon:"■",name:"Chemical
Injection",items:[["Demulsifier","12.5 L/hr","88"],["Corrosion Inh.","5.2 L/hr","76"],["Scale Inh.","4.1
L/hr","80"]]}, {icon:"■■■",name:"Storage Tanks",items:[["Tank 1 Level","72%","72"],["Tank 2
Level","58%","58"],["Tank 3 Level","45%","45"]]}, {icon:"■",name:"Export",items:[["Export Rate","100,000
bpd","85"],["Pipeline Pressure","800 psig","80"],["Meter Factor","1.0002","99"]]} ]; g.innerHTML="";
sections.forEach(function(s,si){ var c='<div class="ctrl-card"><div class="ctrl-head">'+s.icon+"
"+s.name+"</div>"; s.items.forEach(function(it){ var
clr=parseFloat(it[2])>80?"#00e676":parseFloat(it[2])>50?"#ffab40":"#ff5252"; c+=''<div
class="ctrl-row"><span>'+it[0]+'</span><span class="ctrl-val">'+it[1]+'</span></div><div
class="ctrl-bar"><div class="ctrl-fill" style="width: '+it[2]+'%;background: '+clr+' "></div></div>'; });
c+=''</div>";g.innerHTML+=c; }); }

```

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// ===== INIT ===== document.addEventListener("DOMContentLoaded",function(){ renderTrains();
renderControl(); calcSeparator(); calcGas(); calcOil(); calcWater(); calcChemical(); calcStorage();
initAnalyticsCharts(); applyLang(); });

```

■ File: charts.js

```
// Charts module – Enhanced with clearer visuals and accurate data
var chartInstances={}; function
destroyChart(id){if(chartInstances[id]){chartInstances[id].destroy();delete chartInstances[id]} //
Light-mode chart options with improved readability
var cOpts={ responsive:true, maintainAspectRatio:true,
plugins:{ legend:{ labels:{ color:"#4a5568", font:{size:11,family:"Inter,sans-serif",weight:'600'},
padding:16, usePointStyle:true, pointStyleWidth:10 } }, tooltip:{ backgroundColor:"#1a2030",
titleColor:"#00f0ff", bodyColor:"#e0e8f0", borderColor:"rgba(0,240,255,0.2)", borderWidth:1, padding:12,
cornerRadius:10, titleFont:{family:"Inter",weight:'700',size:12}, bodyFont:{family:"JetBrains
Mono",size:11}, displayColors:true, boxPadding:4 } }, scales:{ x:{
ticks:{color:"#4a5568",font:{size:10,family:"Inter",weight:'500'}},
grid:{color:"rgba(0,0,0,.04)",drawBorder:false}, border:{display:false} }, y:{
ticks:{color:"#4a5568",font:{size:10,family:"Inter",weight:'500'}},
grid:{color:"rgba(0,0,0,.05)",drawBorder:false}, border:{display:false} } },
interaction:{intersect:false,mode:'index'}, animation:{duration:800,easing:'easeOutQuart'} }; function
buildSepChart(P,D,L,V){ destroyChart("chartSep"); var ps=[],ds=[],ls=[],areas=[]; for(var
p=50;p<=500;p+=25){ var SG0=0.86,SG=0.75,T=120,K=0.35; var
rho0=SG0*62.4,rhoG=SG*0.0764*(p+14.7)/14.7*(520/(T+460)); var Vm=K*Math.sqrt((rho0-rhoG)/rhoG); ps.push(p+'
psig'); ds.push(parseFloat(Vm.toFixed(3))); ls.push(parseFloat((L*p/P).toFixed(2)));
areas.push(parseFloat((Math.PI/4*Math.pow(D*p/P,2)).toFixed(2))); }
```

```
chartInstances["chartSep"]=new Chart($("#chartSep"),{ type:"line", data:{ labels:ps, datasets:[ {label:"Max
Gas Velocity (ft/s)",data:ds,borderColor:"#0066aa",backgroundColor:"rgba(0,102,170,0.08)",fill:true,tension:
0.4,borderWidth:2.5,pointRadius:2,pointHoverRadius:6,pointBackgroundColor:"#0066aa"}, {label:"Vessel Length
Factor (ft)",data:ls,borderColor:"#0a8a3e",backgroundColor:"rgba(10,138,62,0.08)",fill:true,tension:0.4,bord
erWidth:2.5,pointRadius:2,pointHoverRadius:6,pointBackgroundColor:"#0a8a3e",yAxisID:"y1"} ] },
options:Object.assign({},cOpts,{ scales:Object.assign({},cOpts.scales,{
x:Object.assign({},cOpts.scales.x,{title:{display:true,text:'Operating
Pressure',color:'#8896a6',font:{size:10,weight:'600'}}}),
y:Object.assign({},cOpts.scales.y,{title:{display:true,text:'Gas Velocity
(ft/s)',color:'#0066aa',font:{size:10,weight:'600'}}}), y1:{position:"right",ticks:{color:"#0a8a3e",font:{si
ze:10}},grid:{display:false},border:{display:false},title:{display:true,text:'Length Factor
(ft)',color:'#0a8a3e',font:{size:10,weight:'600'}} } ) } }); } function buildGasChart(Q,Pin,stages,energy){
destroyChart("chartGas"); var ls=[],ps=[],es=[]; for(var q=10;q<=200;q+=10){ ls.push(q+' MMSCFD'); var
hp=q*1e6/86400*144*(Pin+14.7)/(33000)*stages*0.4; ps.push(parseFloat(hp.toFixed(1))); es.push(q*1000); }
chartInstances["chartGas"]=new Chart($("#chartGas"),{ type:"line", data:{ labels:ls, datasets:[
{label:"Compression HP",data:ps,borderColor:"#e68a00",backgroundColor:"rgba(230,138,0,0.08)",fill:true,tensi
on:0.4,borderWidth:2.5,pointRadius:2,pointHoverRadius:6,pointBackgroundColor:"#e68a00"}, {label:"Thermal
Energy (MMBTU/D)",data:es,borderColor:"#0066aa",backgroundColor:"rgba(0,102,170,0.08)",fill:true,tension:0.4
,borderWidth:2.5,pointRadius:2,pointHoverRadius:6,pointBackgroundColor:"#0066aa",yAxisID:"y1"} ] },
options:Object.assign({},cOpts,{ scales:Object.assign({},cOpts.scales,{
x:Object.assign({},cOpts.scales.x,{title:{display:true,text:'Gas Flow
Rate',color:'#8896a6',font:{size:10,weight:'600'}}}),
y:Object.assign({},cOpts.scales.y,{title:{display:true,text:'Compression
HP',color:'#e68a00',font:{size:10,weight:'600'}}}), y1:{position:"right",ticks:{color:"#0066aa",font:{size:1
0}},grid:{display:false},border:{display:false},title:{display:true,text:'Energy
(MMBTU/D)',color:'#0066aa',font:{size:10,weight:'600'}} } ) } }); } function
buildOilChart(Q,BSWin,BSWout,duty){ destroyChart("chartOil"); var ls=[],wr=[],du=[]; for(var
b=1;b<=30;b+=1){ ls.push(b+'%'); wr.push(parseFloat((Q*b/100).toFixed(0))); var
d=Q*5.615*52*0.5*(150-100)/1e6; du.push(parseFloat((d*(b/Math.max(BSWin,1))).toFixed(3))); }
chartInstances["chartOil"]=new Chart($("#chartOil"),{ type:"bar", data:{ labels:ls,
```

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datasets:[ {label:"Water Removed (bpd)",data:wr,backgroundColor:"rgba(0,102,170,0.25)",borderColor:"#0066aa"
,borderWidth:1.5,borderRadius:4,barPercentage:0.7}, {label:"Heater Load Factor",data:du,backgroundColor:"rgb
a(230,138,0,0.25)",borderColor:"#e68a00",borderWidth:1.5,borderRadius:4,barPercentage:0.7} ] },
options:Object.assign({},cOpts,{ scales:Object.assign({},cOpts.scales,{
x:Object.assign({},cOpts.scales.x,{title:{display:true,text:'BSW
%',color:'#8896a6',font:{size:10,weight:'600'}}}),
y:Object.assign({},cOpts.scales.y,{title:{display:true,text:'Volume /
Factor',color:'#8896a6',font:{size:10,weight:'600'}}}) } ) } }); } function
buildWaterChart(Q,OiW,OiWout,eff){ destroyChart("chartWater"); chartInstances["chartWater"]=new
Chart($("#chartWater"),{ type:"doughnut", data:{ labels:["Oil Removed (" +Math.round(OiW-OiWout)+") ppm","Oil
Remaining (" +Math.round(OiWout)+") ppm","Clean Water"], datasets:[{
data:[OiW-OiWout,OiWout,Math.max(0,1000-OiW)], backgroundColor:["#0a8a3e","#c0392b","#0066aa"],
hoverBackgroundColor:["#2ecc71","#e74c3c","#3498db"], borderWidth:2, borderColor:'#fff' } ] }, options:{
responsive:true, cutout:'55%', plugins:{ legend:{ position:'bottom',
labels:{color:"#4a5568",font:{size:11,family:"Inter",weight:'600'},padding:16,usePointStyle:true} },
```

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tooltip:{ backgroundColor:"#1a2030",titleColor:"#00f0ff",bodyColor:"#e0e8f0",
borderColor:"rgba(0,240,255,0.2)",borderWidth:1,padding:12,cornerRadius:10 } },
animation:{animateRotate:true,animateScale:true,duration:1000} } }); } function
buildChemChart(labels,vals,colors){ destroyChart("chartChem"); var bgColors = ['rgba(0,102,170,0.3)','rgba(1
0,138,62,0.3)','rgba(230,138,0,0.3)','rgba(194,24,91,0.3)','rgba(156,39,176,0.3)','rgba(52,152,219,0.3)','rg
ba(243,156,18,0.3)']; var bdColors =
['#0066aa','#0a8a3e','#e68a00','#c2185b','#9c27b0','#3498db','#f39c12']; chartInstances["chartChem"]=new
Chart$("chartChem",{ type:"bar", data:{ labels:labels, datasets:[{ label:"Injection Rate (L/hr)",
data:vals, backgroundColor:bgColors, borderColor:bdColors, borderWidth:1.5,

```

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borderRadius:6, barPercentage:0.65 } } }, options:Object.assign({},cOpts,{ indexAxis:'y', scales:{
x:Object.assign({},cOpts.scales.x,{title:{display:true,text:'Rate
(L/hr)',color:'#8896a6',font:{size:10,weight:'600'}}}),
y:Object.assign({},cOpts.scales.y,{ticks:{font:{size:11,weight:'600'}}}) } } }); } function
initAnalyticsCharts(){ // 1. Efficiency vs Pressure – bell-curve based on real separator data var
ps2=[],ef=[]; for(var p=50;p<=400;p+=20){ ps2.push(p+' psig'); var
e=82+0.03*(p-50)-0.00005*Math.pow(p-200,2); ef.push(parseFloat(Math.min(98,Math.max(70,e)).toFixed(1))); }
new Chart$("chartEffP",{ type:"line", data:{labels:ps2,datasets:[{ label:"Separation Efficiency
%",data:ef,
borderColor:"#0a8a3e",backgroundColor:"rgba(10,138,62,0.08)",fill:true,tension:0.4,borderWidth:2.5,
pointRadius:3,pointHoverRadius:7,pointBackgroundColor:"#0a8a3e",pointBorderColor:"#fff",pointBorderWidth:1.5
}]], options:Object.assign({},cOpts,{ scales:Object.assign({},cOpts.scales,{
x:Object.assign({},cOpts.scales.x,{title:{display:true,text:'Pressure
(psig)',color:'#8896a6',font:{size:10,weight:'600'}}}),
y:{min:65,max:100,ticks:{color:"#4a5568",font:{size:10,callback:function(v){return
v+'%'}}},grid:{color:"rgba(0,0,0,.05)",border:{display:false},title:{display:true,text:'Efficiency
%',color:'#0a8a3e',font:{size:10,weight:'600'}}}) } } }); // 2. GOR vs API var apis=[],gors=[]; for(var
a=15;a<=50;a+=2){ apis.push(a+'° API'); gors.push(parseFloat((200+Math.pow(a-15,1.5)*8).toFixed(0))); } new
Chart$("chartGorApi",{ type:"line", data:{labels:apis,datasets:[{ label:"GOR (SCF/bbl)",data:gors,
borderColor:"#3d2200",backgroundColor:"rgba(61,34,0,0.06)",fill:true,tension:0.4,borderWidth:2.5,
pointRadius:3,pointHoverRadius:7,pointBackgroundColor:"#a05a00",pointBorderColor:"#fff",pointBorderWidth:1.5
}]], options:Object.assign({},cOpts,{ scales:Object.assign({},cOpts.scales,{
x:Object.assign({},cOpts.scales.x,{title:{display:true,text:'API
Gravity',color:'#8896a6',font:{size:10,weight:'600'}}}),
y:Object.assign({},cOpts.scales.y,{title:{display:true,text:'GOR
(SCF/bbl)',color:'#3d2200',font:{size:10,weight:'600'}}}) } } }); // 3. Retention Time vs Flow Rate var
qs=[],rets=[];

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for(var q=5000;q<=80000;q+=5000){ qs.push((q/1000)+'K bpd'); var im=q*0.15898;
rets.push(parseFloat((im>0?(60*1440)/im:0).toFixed(2))); } new Chart$("chartRetQ",{ type:"line",
data:{labels:qs,datasets:[{ label:"Retention Time (min)",data:rets,
borderColor:"#0066aa",backgroundColor:"rgba(0,102,170,0.06)",fill:true,tension:0.4,borderWidth:2.5,
pointRadius:3,pointHoverRadius:7,pointBackgroundColor:"#0066aa",pointBorderColor:"#fff",pointBorderWidth:1.5
}]], options:Object.assign({},cOpts,{ scales:Object.assign({},cOpts.scales,{
x:Object.assign({},cOpts.scales.x,{title:{display:true,text:'Liquid Flow
Rate',color:'#8896a6',font:{size:10,weight:'600'}}}),
y:Object.assign({},cOpts.scales.y,{title:{display:true,text:'Retention Time
(min)',color:'#0066aa',font:{size:10,weight:'600'}}}) } } }); // 4. Chemical Cost – improved pie with
labels new Chart$("chartChemCost",{ type:"pie", data:{ labels:["Demulsifier 25%","Corr. Inh. 15%","Scale
Inh. 12%","H2S Scav. 18%","Biocide 10%","O2 Scav. 8%","PPD 12%"], datasets:[{ data:[25,15,12,18,10,8,12],
backgroundColor:["#0066aa","#0a8a3e","#e68a00","#c2185b","#9c27b0","#3498db","#f39c12"],
hoverBackgroundColor:["#3498db","#2ecc71","#ffc107","#e91e63","#ba68c8","#5dade2","#f5b041"], borderWidth:2,
borderColor:'#fff' } } }, options:{ responsive:true, plugins:{ legend:{ position:'right',
labels:{color:"#4a5568",font:{size:10,family:"Inter",weight:'600'},padding:12,usePointStyle:true} } },
animation:{animateRotate:true,animateScale:true,duration:1000} } }); }

```

■ File: features.js

```
"use strict"; // ===== NAV HISTORY ===== var navHistory=[],navIdx=-1,navLock=false; function histPush(t){if(
navLock)return;if(navIdx<navHistory.length-1)navHistory=navHistory.slice(0,navIdx+1);navHistory.push(t);navI
dx=navHistory.length-1} function
histBack(){if(navIdx>0){navLock=true;navIdx--;switchTab(navHistory[navIdx]);navLock=false}} function
histFwd(){if(navIdx<navHistory.length-1){navLock=true;navIdx++;switchTab(navHistory[navIdx]);navLock=false}}
// patch switchTab var _origSwitch=switchTab; switchTab=function(t){_origSwitch(t);histPush(t)}; // =====
SAVE PANEL AS IMAGE ===== function savePanel(id){ var el=document.getElementById(id);if(!el)return;
if(typeof html2canvas===undefined){showToast('html2canvas not loaded');return} showToast('Processing
image... please wait'); // Temporary fix for scrolling issue with html2canvas window.scrollTo(0,0);
html2canvas(el,{backgroundColor:'#1a2030', scale:2, useCORS:true, allowTaint:false}).then(function(c){ var
a=document.createElement('a'); a.download='Hammar_'+id+'_'+Date.now()+'.png';
a.href=c.toDataURL('image/png'); a.click(); showToast('Saved as image!'); }).catch(function(err){
showToast('Error saving image'); console.error(err); }); } function showToast(msg){var t=document.getElement
ById('toast');if(!t)return;t.textContent=msg;t.classList.add('show');setTimeout(function(){t.classList.remove
('show')},2500)} // ===== ENTER KEY ===== document.addEventListener('keydown',function(e){
if(e.key!=='Enter')return; var active=document.querySelector('.tab-panel.active');if(!active)return; var
id=active.id; if(id==='tab-separator')calcSeparator(); else if(id==='tab-gas')calcGas(); else
if(id==='tab-oil')calcOil(); else if(id==='tab-water')calcWater(); else
if(id==='tab-chemical')calcChemical(); else if(id==='tab-storage')calcStorage(); }); // ===== EQUATIONS
PANEL ===== var equations=[ {name:'Retention Time',std:'API 12J',formula:'t = V_liquid /
Q_liquid',desc:'Liquid retention time in separator vessel',
vars:[['V_liquid','60','m³'],['Q_liquid','5000','bpd']], calc:function(v){var Qm3=v[1]*0.15898/1440;return
Qm3>0?(v[0]/Qm3).toFixed(2)+' min':'N/A'}}, {name:'Stokes Law Settling',std:'API 12J',formula:'V_t = (Δp × g
× d²) / (18 × μ)',desc:'Terminal settling velocity for droplet separation', vars:[['Δp
(lb/ft³)','12','lb/ft³'],['d (micron)','150','μm'],['μ (cp)','3.5','cp']], calc:function(v){var
df=v[1]*1e-6*3.281;var vt=(v[0]*32.174*df*df)/(18*v[2]*6.72e-4);return (vt*12).toFixed(4)+' in/s'}},
{name:'Souders-Brown',std:'API 12J / GPSA',formula:'V_max = K × √((p_L - p_G) / p_G)',desc:'Maximum allowable
gas velocity in separator', vars:[['K factor','0.35','-'],['p_L (lb/ft³)','53.5','lb/ft³'],['p_G
(lb/ft³)','0.5','lb/ft³']], calc:function(v){return (v[0]*Math.sqrt((v[1]-v[2])/v[2])).toFixed(3)+' ft/s'}},
{name:'Oil Density from API',std:'API / ASTM D287',formula:'SG = 141.5 / (API + 131.5)',desc:'Specific
gravity from API gravity', vars:[['API Gravity','32','°API']], calc:function(v){var
sg=141.5/(v[0]+131.5);return 'SG = '+sg.toFixed(4)+' | ρ = '+ (sg*62.4).toFixed(2)+' lb/ft³'}}, {name:'GOR
Calculation',std:'Field Units',formula:'GOR = (Q_gas × 10■) / Q_oil',desc:'Gas-Oil Ratio in standard cubic
feet per barrel', vars:[['Q_gas (MMSCFD)','50','MMSCFD'],['Q_oil (bpd)','25000','bpd']],
calc:function(v){return v[1]>0?((v[0]*1e6)/v[1]).toFixed(1)+' SCF/bbl':'N/A'}},
```

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{name:'Water Cut',std:'Field Units',formula:'WC = Q_water / (Q_oil + Q_water) × 100',desc:'Water cut
percentage', vars:[['Q_water (bpd)','10000','bpd'],['Q_oil (bpd)','25000','bpd']], calc:function(v){var
t=v[0]+v[1];return t>0?((v[0]/t)*100).toFixed(2)+' %':'N/A'}}, {name:'Darcy Flow',std:'API /
Darcy',formula:'Q = (k × A × ΔP) / (μ × L)',desc:'Flow rate through porous media', vars:[['k
(mD)','100','mD'],['A (ft²)','10','ft²'],['ΔP (psi)','500','psi'],['μ (cp)','3.5','cp'],['L
(ft)','1000','ft']], calc:function(v){var q=(v[0]*v[1]*v[2])/v[3]*v[4];return q.toFixed(4)+' bbl/day
(simplified)}}, {name:'Heater Duty',std:'API / ASTM',formula:'Q = ■ × Cp × ΔT',desc:'Heat duty for oil
heating', vars:[['Q_oil (bpd)','25000','bpd'],['p (lb/ft³)','53.5','lb/ft³'],['ΔT (°F)','50','°F']],
calc:function(v){var m=v[0]*5.615*v[1]/24;var duty=m*0.5*v[2]/1e6;return duty.toFixed(3)+' MMBTU/hr'}},
{name:'Chemical Dose Rate',std:'Field Units',formula:'Rate = Q_L × 158.98 × dose / 10■',desc:'Chemical
injection rate from ppm dosage', vars:[['Q_liquid (bpd)','35000','bpd'],['Dose (ppm)','15','ppm']],
calc:function(v){var ld=v[0]*158.98*v[1]/1e6;return ld.toFixed(3)+' L/day | +(ld/24).toFixed(4)+' L/hr'}},
{name:'Z-Factor (Approx)',std:'Hall-Yarborough',formula:'Z = f(T_pr, P_pr)',desc:'Gas compressibility factor
approximation', vars:[['SG_gas','0.75','-'],['P (psig)','150','psig'],['T (°F)','120','°F']],
calc:function(v){var Tpc=168+325*v[0]-12.5*v[0]*v[0];var Ppc=677+15*v[0]-37.5*v[0]*v[0];var
Tpr=(v[2]+460)/Tpc;var Ppr=(v[1]+14.7)/Ppc;var Z=1-(3.53*Ppr)/(Math.pow(10,0.9813*Tpr))+0.274*Ppr*Ppr/(Math.
pow(10,0.8157*Tpr));Z=Math.max(0.3,Math.min(1.2,Z));return 'Z = '+Z.toFixed(4)}}, {name:'Compression
Ratio',std:'GPSA / API 618',formula:'CR = P_discharge / P_suction',desc:'Overall and per-stage compression
ratio', vars:[['P_suction (psig)','150','psig'],['P_discharge (psig)','1000','psig']], calc:function(v){var
cr=(v[1]+14.7)/(v[0]+14.7);var n=Math.max(1,Math.ceil(Math.log(cr)/Math.log(3.5)));return 'CR =
'+cr.toFixed(2)+' : 1 | Stages: '+n+' | Per stage: '+Math.pow(cr,1/n).toFixed(2)+' : 1'}}, {name:'Skim Tank
Volume',std:'API 421',formula:'V = Q × t_ret',desc:'Required skim tank volume for water treatment',
vars:[['Q_water (bpd)','10000','bpd'],['t_ret (min)','15','min']], calc:function(v){var
qf=v[0]*5.615/1440;var vol=qf*v[1];return vol.toFixed(2)+' ft³ | +(vol*0.178107).toFixed(2)+' bbl'}} ];
function renderEquations(){ var g=document.getElementById('eqGrid');if(!g)return;g.innerHTML='';
equations.forEach(function(eq,ei){ var h='<div class="eq-card"><h3>'+eq.name+' <span
style="font-size:10px;color:#8896a6;font-weight:400">'+eq.std+'</span></h3>'; h+=''<div
class="eq-formula">'+eq.formula+'</div>'; h+=''<div class="eq-desc">'+eq.desc+'</div>'; h+=''<div
class="eq-inputs">'; eq.vars.forEach(function(v,vi){ h+=''<label>'+v[0]+'<input type="number"
id="eq-'+ei+'-'+vi+'" value="'+v[1]+'"" step="any"><span>'+v[2]+'</span></label>'; }); h+=''</div>';
```

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h+=''<button class="eq-calc-btn" onclick="calcEq('+ei+')">Calculate</button>; h+=''<div class="eq-result"
id="eq-res-'+ei+'"></div></div>; g.innerHTML+=h; }); } function calcEq(i){ var eq=equations[i];var
vals=[];
eq.vars.forEach(function(v,vi){vals.push(parseFloat(document.getElementById('eq-'+i+'-'+vi).value)||0)});
document.getElementById('eq-res-'+i).textContent=eq.calc(vals); } // ===== DETAILED PROCESS DIAGRAM =====
var equipInfo={ wellhead:{name:'Wellhead / Christmas Tree',desc:'Production wellhead assembly including
tubing hanger, casing head, and choke valve. Controls well flow and provides primary pressure reduction.
Typical pressure: 1500-3000 psig.'}, manifold:{name:'Production Manifold',desc:'Collects production from
multiple wells and distributes to processing trains. Includes test headers, emergency shutdown valves
(ESDVs), and flow measurement.'}, hpsep:{name:'HP Separator (1st Stage)',desc:'High-pressure three-phase
separator. Separates gas, oil, and water at operating pressure 100-200 psig. Designed per API 12J with
Souders-Brown and Stokes criteria.'}, lpsep:{name:'LP Separator (2nd Stage)',desc:'Low-pressure three-phase
separator operating at 30-75 psig. Further separates dissolved gas and provides additional water-oil
separation. Longer retention time.'}, gasplant:{name:'Gas Treatment Plant',desc:'Includes gas sweetening
(amine unit), dehydration (TEG), and compression. Removes H2S, CO2, and water vapor to meet sales gas
specifications per GPSA standards.'}, heater:{name:'Heater Treater / Steam',desc:'Provides heat to reduce
oil viscosity and break emulsions. Typical outlet temperature 140-160°F. Duty calculated per API standards.
May use direct-fired or steam coils.'}, desalter:{name:'Electrostatic Desalter',desc:'Removes salt from
crude oil using wash water and high-voltage electrostatic field (15-25 kV). Reduces salt content to <10 PTB
for export specifications.'}, demulinj:{name:'Demulsifier Injection',desc:'Chemical injection point for
demulsifier/emulsion breaker. Typical dose 5-30 ppm. Breaks water-in-oil emulsions to facilitate separation
in downstream vessels.'}, corrinj:{name:'Corrosion Inhibitor Injection',desc:'Injects corrosion inhibitor to
protect piping and vessels. Film-forming amines typical at 5-15 ppm. Critical for high H2S/CO2
environments.'}, watertreat:{name:'Water Treatment (WSEP/CPI)',desc:'Produced water treatment including
skim tanks, corrugated plate interceptors (CPI), and hydrocyclones. Reduces oil-in-water to <40 ppm for
disposal/reinjection.'}, storage:{name:'Crude Storage Tanks',desc:'Atmospheric storage tanks per API 650.
Fixed or floating roof design. Provides buffer between production and export. Typical capacity:
500,000-1,000,000 bbl.'}, export:{name:'Export / Loading',desc:'Export pumps, custody transfer metering (per
API MPMS), and loading facilities. Includes LACT units with BS&W analyzers and fiscal metering.'} };

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function renderDetailDiagram(){ var w=document.getElementById('detailDiagramWrap');if(!w)return; var
svg='<svg viewBox="0 0 1200 600" xmlns="http://www.w3.org/2000/svg"
style="width:100%;height:auto;min-width:800px">; // Background
svg+=''<rect width="1200" height="600"
fill="#f8fafb" rx="12"/>; svg+=''<text x="600" y="30" text-anchor="middle" fill="#1a2030" font-size="14"
font-weight="700" font-family="Inter,sans-serif">HAMMAR IPF – Complete Process Flow Diagram (Wellhead →
Export)</text>; // Phase color legend
svg+=''<g transform="translate(30,560)"><rect width="14" height="8"
rx="2" fill="#3d2200"/><text x="20" y="8" fill="#4a5568" font-size="9" font-family="Inter">Oil</text>;
svg+=''<rect x="60" width="14" height="8" rx="2" fill="#0a8a3e"/><text x="80" y="8" fill="#4a5568"
font-size="9" font-family="Inter">Gas</text>; svg+=''<rect x="120" width="14" height="8" rx="2"
fill="#0066aa"/><text x="140" y="8" fill="#4a5568" font-size="9" font-family="Inter">Water</text>;
svg+=''<rect x="190" width="14" height="8" rx="2" fill="#c2185b"/><text x="210" y="8" fill="#4a5568"
font-size="9" font-family="Inter">Chemical</text></g>; function node(x,y,w,h,color,key,label,icon){ return
'<g class="svg-node" style="cursor:pointer" onclick="showEquipInfo(\'+key+\',event)">'+<rect x="'+x+'
y="'+y+' width="'+w+' height="'+h+' rx="10" fill="#fff" stroke="'+color+' stroke-width="2"/>'+<text
x="'+(x+w/2)+' y="'+(y+h/2-6)+' text-anchor="middle" fill="'+color+' font-size="20">'+icon+'</text>'+
<text x="'+(x+w/2)+' y="'+(y+h/2+12)+' text-anchor="middle" fill="#1a2030" font-size="9"
font-weight="700" font-family="Inter,sans-serif">'+label+'</text></g>; } function
arrow(x1,y1,x2,y2,color,dash){ var d=dash? ' stroke-dasharray="6,3"':''; return '<line x1="'+x1+'
y1="'+y1+' x2="'+x2+' y2="'+y2+' stroke="'+color+' stroke-width="3"+d+'>'+<animate
attributeName="stroke-dashoffset" from="9" to="0" dur="1s" repeatCount="indefinite"/></line>; } // Nodes -
row 1 (main process)
svg+=node(30,200,90,70,'#3d2200','wellhead','WELLHEAD','');
svg+=arrow(120,235,160,235,'#3d2200',true); svg+=node(165,200,90,70,'#e68a00','manifold','MANIFOLD','');
svg+=arrow(255,235,295,235,'#3d2200',true); svg+=node(300,195,130,80,'#0a8a3e','hpsep','HP
SEPARATOR','');
svg+=arrow(430,235,470,235,'#3d2200',true);
svg+=node(475,195,130,80,'#0a8a3e','lpsep','LP SEPARATOR','');
svg+=arrow(605,235,645,235,'#3d2200',true); svg+=node(650,200,90,70,'#e68a00','heater','HEATER','');
svg+=arrow(740,235,780,235,'#3d2200',true); svg+=node(785,200,100,70,'#9c27b0','desalter','DESALTER','');
svg+=arrow(885,235,925,235,'#3d2200',true); svg+=node(930,200,90,70,'#e68a00','storage','STORAGE','');
svg+=arrow(1020,235,1060,235,'#3d2200',false); svg+=node(1065,200,100,70,'#0a8a3e','export','EXPORT','');
// Gas treatment (top)
svg+=arrow(365,195,365,110,'#0a8a3e',true);
svg+=node(310,50,110,60,'#0a8a3e','gasplant','GAS TREATMENT','');
svg+=arrow(540,195,540,110,'#0a8a3e',true); svg+=''<line x1="420" y1="80" x2="540" y2="80" stroke="#0a8a3e"
stroke-width="2" stroke-dasharray="4,3"/>; // Water treatment (bottom)
svg+=arrow(540,275,540,370,'#0066aa',true); svg+=node(490,375,110,60,'#0066aa','watertreat','WATER
TREAT',''); // Chemical injection
svg+=node(300,420,90,50,'#c2185b','demulinj','DEMULSIFIER','');
svg+=''<line x1="345" y1="420" x2="345" y2="275" stroke="#c2185b" stroke-width="1.5"
stroke-dasharray="4,3"><animate attributeName="stroke-dashoffset" from="7" to="0" dur="1.2s"
repeatCount="indefinite"/></line>; svg+=node(680,420,90,50,'#c2185b','corrinj','CORR. INH.','');
svg+=''<line x1="725" y1="420" x2="725" y2="270" stroke="#c2185b" stroke-width="1.5"

```

```
stroke-dasharray="4,3"><animate attributeName="stroke-dashoffset" from="7" to="0" dur="1.2s"
repeatCount="indefinite"/></line>'; svg+=''</svg>'; w.innerHTML=svg;
```

```
} function showEquipInfo(key,evt){ var info=equipInfo[key];if(!info)return; var
popup=document.getElementById('equipPopup');if(!popup)return;
popup.innerHTML='<h4>'+info.name+'</h4><p>'+info.desc+'</p>'; popup.classList.add('show'); var
rect=evt.target.closest('.svg-node').getBoundingClientRect(); var
wrap=document.getElementById('tab-diagram').getBoundingClientRect();
popup.style.left=Math.min(rect.left-wrap.left,wrap.width-320)+'px';
popup.style.top=(rect.bottom-wrap.top+8)+'px'; setTimeout(function(){popup.classList.remove('show')},5000);
document.addEventListener('click',function h(e){if(!popup.contains(e.target)){popup.classList.remove('show')}
;document.removeEventListener('click',h)}},{once:false}); } // ===== ENHANCED CHEM DONUT ===== var
_origCalcChem=calcChemical; calcChemical=function(){ _origCalcChem(); // Build additives donut with field
units var Ql=parseFloat(document.getElementById('chem-Ql').value)||0; var chems=[
{name:'Demulsifier',id:'chem-demul',color:'#0066aa'}, {name:'Corrosion
Inh.',id:'chem-corr',color:'#0a8a3e'}, {name:'Scale Inh.',id:'chem-scale',color:'#e68a00'}, {name:'H2S
Scavenger',id:'chem-h2s',color:'#c2185b'}, {name:'Biocide',id:'chem-bio',color:'#9c27b0'}, {name:'O2
Scavenger',id:'chem-o2',color:'#3498db'}, {name:'PPD',id:'chem-ppd',color:'#f39c12'} ]; var legend='<div
class="additives-legend">'; chems.forEach(function(c){ var
dose=parseFloat(document.getElementById(c.id).value)||0; var galDay=Ql*158.98*dose/1e6*0.264172;
legend+='<div class="additives-legend-item"><div class="additives-legend-dot"
style="background: '+c.color+' "></div>'+c.name+' ('+dose+' ppm)<span
class="additives-legend-val">'+galDay.toFixed(2)+' gal/day</span></div>'; }); legend+='</div>'; // Add
legend after chart var chartBox=document.getElementById('chartChem'); if(chartBox){ var
wrap=chartBox.closest('.chart-box'); var existing=wrap.querySelector('.additives-chart-wrap');
if(existing)existing.remove(); var div=document.createElement('div');div.className='additives-chart-wrap';
div.innerHTML=legend;wrap.appendChild(div); } }; // ===== SIMULATION MODE ===== var simActive = false; var
simInterval = null; function toggleSimulation() { simActive = !simActive; var btn =
document.getElementById('simBtn'); var t = i18n[currentLang]; if (simActive) { btn.classList.add('active');
btn.textContent = t.sim_on; simInterval = setInterval(runSimulationStep, 2000);
```

```
showToast(currentLang === 'ar' ? '■■■■ ■■■ ■■■■■■■■■■' : 'Simulation Mode Started'); } else {
btn.classList.remove('active'); btn.textContent = t.sim_off; clearInterval(simInterval);
showToast(currentLang === 'ar' ? '■■■■■ ■■■ ■■■■■■■■■■' : 'Simulation Mode Stopped'); } } function
runSimulationStep() { // Slightly fluctuate inputs var inputs = ['sep-P', 'sep-T', 'sep-Qo', 'sep-Qg'];
inputs.forEach(function(id) { var el = document.getElementById(id); if (!el) return; var val =
parseFloat(el.value); var change = (Math.random() - 0.5) * (val * 0.02); // 2% fluctuation el.value = (val +
change).toFixed(2); }); // Re-calculate everything calcSeparator(); calcGas(); calcOil(); calcWater(); //
Check for alarms checkAlarms(); // Update PID if active if (simActive) simulatePIDLevel(); } // ===== ALARMS
SYSTEM ===== var alarms = []; function toggleAlarms() {
document.getElementById('alarmPanel').classList.toggle('minimized'); } function checkAlarms() { var p =
parseFloat(document.getElementById('sep-P').value); var q =
parseFloat(document.getElementById('sep-Qo').value); var newAlarms = []; if (p > 180) newAlarms.push({ msg:
'High Pressure Alert: > 180 psig', level: 'critical' }); if (p < 100) newAlarms.push({ msg: 'Low Pressure
Warning: < 100 psig', level: 'warning' }); if (q > 40000) newAlarms.push({ msg: 'High Production Load: > 40k
bpd', level: 'warning' }); // Update UI renderAlarms(newAlarms); } function renderAlarms(currentAlarms) {
var list = document.getElementById('alarmList'); var count = document.getElementById('alarmCount');
list.innerHTML = ''; currentAlarms.forEach(function(a) { var item = document.createElement('div');
```

```
item.className = 'alarm-item ' + (a.level === 'warning' ? 'warning' : ''); item.innerHTML = '<span
class="alarm-time">' + new Date().toLocaleTimeString() + '</span>' + '<strong>' + a.msg + '</strong>';
list.appendChild(item); }); count.textContent = currentAlarms.length; // If many alarms, maybe don't
auto-open but pulse the button if (currentAlarms.length > 0 &&
document.getElementById('alarmPanel').classList.contains('minimized')) {
document.getElementById('alarmToggle').style.animation = 'glow 1s infinite'; } else {
document.getElementById('alarmToggle').style.animation = 'none'; } } // ===== PID CONTROL LOGIC ===== var
pidLevel = 50; var pidTarget = 50; var pidValve = 50; function updatePID() { pidValve =
document.getElementById('pidValveSlider').value; document.getElementById('pidValveVal').textContent =
pidValve + '%'; // Adjust level based on valve (simple inverse relationship for demo) // In reality, this
would be a differential equation } function simulatePIDLevel() { // Simple simulation: inflow is constant,
outflow depends on valve var inflow = 5; var outflow = pidValve / 10; pidLevel += (inflow - outflow);
pidLevel = Math.max(0, Math.min(100, pidLevel)); // Update UI
document.getElementById('pidLevelRect').setAttribute('height', (1.3 * pidLevel));
document.getElementById('pidLevelRect').setAttribute('y', 140 - (1.3 * pidLevel));
document.getElementById('pidLevelText').textContent = Math.round(pidLevel) + '%'; var ret = (pidLevel / 100)
* 6; // Mapping level to retention time document.getElementById('pidRetentionVal').textContent =
ret.toFixed(2); // Animate 3D Isometric View if active var iso = document.querySelector('.isometric-sep
svg'); var liq = document.getElementById('isoLiquid'); if(iso && liq) { var scale = 1 + (Math.sin(Date.now()
```

```

/ 500) * 0.01); iso.style.transform = 'scale(' + scale + ') rotateX(5deg)'; // Animate liquid height in 3D
var h = (pidLevel / 100) * 70; // Map level to SVG height liq.setAttribute('d', 'M45,' + (130 - h) + ' L155,'
+ (130 - h) + ' L155,130 L45,130 Z'); var color = pidLevel > 80 ? '#ff5252' : pidLevel < 20 ? '#ffab40' :
'#00f0ff'; iso.querySelector('path[stroke-width="2"]').setAttribute('stroke', color); } } function autoPID()
{ showToast('Auto PID: Adjusting Valve to reach 50%');

```

```

var interval = setInterval(function() { if (pidLevel > 51) pidValve = Math.min(100, parseInt(pidValve) + 2);
else if (pidLevel < 49) pidValve = Math.max(0, parseInt(pidValve) - 2); else { clearInterval(interval);
showToast('Level Stabilized at 50%'); } document.getElementById('pidValveSlider').value = pidValve;
updatePID(); }, 200); } // ===== PDF REPORT EXPORT ===== function exportPDF() { showToast('Generating Daily
Production Report...'); var reportWindow = window.open('', '_blank'); var content =
document.getElementById('tab-overview').innerHTML; var styles = document.head.innerHTML;
reportWindow.document.write('<html><head>' + styles +
'<style>body{background:#fff;padding:40px;}.panel-nav,.save-btn{display:none;}</style></head><body>');
reportWindow.document.write('<h1>Hammar IPF - Daily Production Report</h1>');
reportWindow.document.write('<p>Generated on: ' + new Date().toLocaleString() + '</p>');
reportWindow.document.write(content); reportWindow.document.write('</body></html>'); setTimeout(function() {
reportWindow.print(); // reportWindow.close(); }, 1000); } // ===== THEME TOGGLE ===== function
toggleTheme() { document.body.classList.toggle('dark-mode'); var isDark =
document.body.classList.contains('dark-mode'); localStorage.setItem('hammarTheme', isDark ? 'dark' :
'light'); showToast(isDark ? 'Dark Mode Active' : 'Light Mode Active'); } // ===== VOICE CONTROL
(EXPERIMENTAL) ===== var voiceActive = false; var recognition = null; function toggleVoice() { if
(!('webkitSpeechRecognition' in window)) { showToast('Voice control not supported in this browser'); return;
} voiceActive = !voiceActive; var btn = document.getElementById('voiceBtn'); if (voiceActive) {
btn.classList.add('active'); recognition = new webkitSpeechRecognition(); recognition.continuous = true;
recognition.lang = currentLang === 'ar' ? 'ar-SA' : 'en-US'; recognition.onresult = function(event) {

```

```

var last = event.results.length - 1; var command = event.results[last][0].transcript.toLowerCase();
handleVoiceCommand(command); }; recognition.start(); showToast('Voice Command: Listening...'); } else {
btn.classList.remove('active'); if (recognition) recognition.stop(); showToast('Voice Command: OFF'); } }
function handleVoiceCommand(cmd) { console.log('Voice Command:', cmd); var t = i18n[currentLang]; if
(cmd.includes('overview') || cmd.includes('■■■■■ ■■■■■')) switchTab('overview'); if (cmd.includes('calc') ||
cmd.includes('■■■■■')) switchTab('separator'); if (cmd.includes('report') || cmd.includes('■■■■■'))
exportPDF(); if (cmd.includes('simulation') || cmd.includes('■■■■■')) toggleSimulation(); // Advanced
Simulation Commands if (cmd.includes('set pressure to') || cmd.includes('■■■ ■■■■■ ■■■')) { var num =
cmd.match(/\d+/); if(num) { document.getElementById('sep-P').value = num[0]; calcSeparator();
showToast('Pressure set to ' + num[0] + ' psig'); } } if (cmd.includes('emergency stop') ||
cmd.includes('■■■■■ ■■■■■')) { simActive = true; toggleSimulation(); // This will turn it off if active
showToast('EMERGENCY STOP EXECUTED', 'critical'); } } // ===== ROLE BASED UI ===== function applyRole(role) {
console.log("Applying Role:", role); var sel = document.getElementById('roleSelect'); if(sel) sel.value =
role; if(role === "Operator") { // Disable some sensitive inputs for Operators
document.querySelectorAll('.edit-cell, .sts-select').forEach(el => { if(!el.id.includes('pid')) el.disabled
= true; }); showToast('Logged in as Operator (Read-Only Mode)'); } else if(role === "Administrator") {
showToast('Administrator Mode: Full Control Granted'); } } // ===== DATA HISTORY LOGGING ===== function
logProduction() { var p = val("sep-P"); var q = val("sep-Qo");

```

```

var log = { time: new Date().toISOString(), pressure: p, oil: q, user: 'Engineer' }; var history =
JSON.parse(localStorage.getItem('hammarHistory') || '[]'); history.push(log); if (history.length > 100)
history.shift(); // Keep last 100 localStorage.setItem('hammarHistory', JSON.stringify(history)); } // =====
INIT NEW FEATURES ===== document.addEventListener('DOMContentLoaded', function() { histPush('overview');
renderEquations(); renderDetailDiagram(); // Minimize alarms by default
document.getElementById('alarmPanel').classList.add('minimized'); // Load Role if exists var savedRole =
sessionStorage.getItem('hammar_role'); if(savedRole) applyRole(savedRole); // Load Theme if
(localStorage.getItem('hammarTheme') === 'dark') { document.body.classList.add('dark-mode'); } // Auto-log
every 5 minutes if simulation is on setInterval(function() { if (simActive) logProduction(); }, 300000); });

```

■ File: login.js

```
"use strict"; (function(){ var loginScreen = document.createElement('div'); loginScreen.id = 'loginScreen';
loginScreen.innerHTML = ` <div class="login-bg"> <div class="login-rings"> <div class="login-ring
ring1"></div> <div class="login-ring ring2"></div> </div> <div class="login-particles"
id="loginParticles"></div> <div class="login-card" id="loginCard"> <!-- Left side: Credits --> <div
class="login-left"> <div class="login-credits-wrap">  <div class="credits-title">Academic
Project</div> <div class="credits-text"> This project was developed and modeled to simulate the Hammar
station by Petroleum Engineering students at Al-Maaqal University, under the direct supervision of:
<strong>"Dr. Eng. Saher Adel & Dr. Mahmoud Badawy"</strong> </div> </div> </div> <!-- Right side: Form -->
<div class="login-right"> <div class="login-logo-wrap"> <!-- EMBEDDED HIGH-FIDELITY SVG LOGO --> <div
class="main-logo station-logo-crop"> <svg viewBox="0 0 100 100" style="width:100%;height:100%;"> <defs>
<linearGradient id="shieldGrad" x1="0" y1="0" x2="1" y2="1"> <stop offset="0%" stop-color="#00f0ff" /> <stop
offset="100%" stop-color="#00e676" /> </linearGradient> </defs> <path d="M50,5 L90,25 L90,75 L50,95 L10,75
L10,25 Z" fill="none" stroke="url(#shieldGrad)" stroke-width="3" /> <rect x="40" y="45" width="8"
height="35" fill="#00f0ff" opacity="0.8" /> <rect x="52" y="35" width="8" height="45" fill="#00e676"
opacity="0.8" /> <path d="M30,40 Q50,20 70,40" fill="none" stroke="#ffc107" stroke-width="2" /> <circle
cx="50" cy="55" r="10" fill="#1a2030" stroke="#00f0ff" stroke-width="1" /> <path d="M50,50 Q55,55 50,62
Q45,55 50,50" fill="#ffab40" /> </svg> </div> <h2 class="login-title">HAMMAR IPF</h2> <p
class="login-subtitle">Smart SCADA Digital Twin</p> </div> <div class="login-field"> <label
class="login-label">ACCESS ROLE</label> <div class="login-input-wrap"> <div class="login-lock-icon">■</div>
<select class="login-input" id="loginRole"
style="appearance:none;padding-right:30px;background:transparent;cursor:pointer;"> <option
value="Operator">OPERATOR</option> <option value="Engineer" selected>ENGINEER</option> <option
value="Administrator">ADMINISTRATOR</option>

</select> </div> </div> <div class="login-field"> <label class="login-label">SECURITY KEY</label> <div
class="login-input-wrap"> <div class="login-lock-icon">■</div> <input type="password" class="login-input"
id="loginPass" placeholder="....." maxlength="10"> <button class="login-eye" id="loginEye"
type="button">■■■</button> </div> </div> <button class="login-btn" id="loginBtn" type="button"> <span
class="login-btn-text">INITIALIZE SYSTEM</span> <span class="login-btn-arrow">→</span> </button> <div
class="login-error" id="loginError"></div> </div> </div> `; document.body.appendChild(loginScreen);
// Particles var pWrap = document.getElementById('loginParticles'); for(var i=0; i<40; i++){ var p =
document.createElement('div'); p.className = 'login-particle'; var size = Math.random()*3 + 1; p.style.width =
size+'px'; p.style.height = size+'px'; p.style.left = Math.random()*100+'%'; p.style.top =
Math.random()*100+'%'; p.style.setProperty('--dur', (Math.random()*4+3)+'s'); p.style.setProperty('--delay',
(Math.random()*5)+'s'); p.style.background = Math.random()>0.5 ? '#00f0ff' : '#00e676';
pWrap.appendChild(p); } var pass = document.getElementById('loginPass'); var eye =
document.getElementById('loginEye'); var btn = document.getElementById('loginBtn'); var err =
document.getElementById('loginError'); var card = document.getElementById('loginCard'); eye.onclick =
function(){ pass.type = pass.type === 'password' ? 'text' : 'password'; eye.textContent = pass.type ===
'password' ? '■■■' : '■'; }; function doLogin(){ var val = pass.value; var role =
document.getElementById('loginRole').value; if(val === "770077"){ err.textContent = "ACCESS GRANTED";
err.className = "login-error show ok"; card.classList.add('success'); sessionStorage.setItem('hammar_auth',
'true'); sessionStorage.setItem('hammar_role', role);

setTimeout(function(){ loginScreen.classList.add('fadeout'); setTimeout(function(){ loginScreen.remove();
if(window.applyRole) window.applyRole(role); }, 600); }, 800); } else { err.textContent = "INVALID SECURITY
KEY"; err.className = "login-error show"; card.classList.add('shake'); pass.value = "";
setTimeout(function(){ card.classList.remove('shake'); }, 500); } } btn.onclick = doLogin; pass.onkeydown =
function(e){ if(e.key === "Enter") doLogin(); }; if(sessionStorage.getItem('hammar_auth') === 'true'){
loginScreen.remove(); } }());
```

■ File: app.js

```
// ===== Hammar IPF - Smart SCADA Dashboard ===== // Interactive GOSP Dashboard Logic (function () { "use
strict"; // ===== INITIAL DATA ===== const initialData = [ { id: "TRAIN 1", status: "ON", inlet: 31731.52,
oil: 22513.10, water: 9218.42, gas: 64.8, sepVol: 60.0 }, { id: "TRAIN 2", status: "ON", inlet: 46508.43,
oil: 37796.91, water: 8711.52, gas: 36.1, sepVol: 60.0 }, { id: "TRAIN 3", status: "ON", inlet: 51067.11,
oil: 39528.68, water: 11538.43, gas: 35.9, sepVol: 60.0 }, { id: "TRAIN 4", status: "STOP", inlet: 0.0, oil:
0.0, water: 0.0, gas: 0.0, sepVol: 60.0 }, ]; // Deep clone for live editing let trainData =
JSON.parse(JSON.stringify(initialData)); // ===== HEADER CLOCK ===== function updateClock() { const now =
new Date(); const el = document.getElementById("headerTime"); if (el) { el.textContent =
now.toLocaleString("en-GB", { day: "2-digit", month: "short", year: "numeric", hour: "2-digit", minute:
"2-digit", second: "2-digit", }); } } setInterval(updateClock, 1000); updateClock(); // ===== FORMATTING
===== function fmt(n) { return n.toLocaleString("en-US", { minimumFractionDigits: 2, maximumFractionDigits:
2 }); } // ===== ENGINEERING CALCULATIONS ===== function calcResults(d) { const waterCut = d.inlet > 0 ?
(d.water / d.inlet) * 100 : 0; const gor = d.oil > 0 ? (d.gas * 1000000) / d.oil : 0; const inletM3 = d.inlet
* 0.15898; const retTime = inletM3 > 0 ? (d.sepVol * 1440) / inletM3 : 0; return { waterCut, gor, inletM3,
retTime }; } function calcGlobals() { const totInlet = trainData.reduce((s, d) => s + d.inlet, 0); const
totOil = trainData.reduce((s, d) => s + d.oil, 0); const totGas = trainData.reduce((s, d) => s + d.gas, 0);
const demulRate = ((totInlet * 158.98) * (15 / 1000000)) / 24; const gasEnergy = totGas * 1000; return {
totInlet, totOil, totGas, demulRate, gasEnergy }; } // ===== RENDER INPUT TABLE ===== function
renderInputTable() { const tbody = document.getElementById("inputTableBody"); tbody.innerHTML = "";
trainData.forEach((train, idx) => {
```

```
const tr = document.createElement("tr"); // Train ID (read-only) const tdId = document.createElement("td");
tdId.innerHTML = `<span style="color:#00f0ff;font-weight:700;">${train.id}</span>`; tr.appendChild(tdId); //
Status (dropdown) const tdStatus = document.createElement("td"); const sel =
document.createElement("select"); sel.className = "status-select"; sel.innerHTML = `<option value="ON"
${train.status==="ON"? "selected":""}>ON</option><option value="STOP"
${train.status==="STOP"? "selected":""}>STOP</option>`; sel.addEventListener("change", function () {
trainData[idx].status = this.value; if (this.value === "STOP") { trainData[idx].inlet = 0;
trainData[idx].oil = 0; trainData[idx].water = 0; trainData[idx].gas = 0; } renderInputTable();
renderResultsTable(); renderKPIs(); }); tdStatus.appendChild(sel); tr.appendChild(tdStatus); // Editable
numeric fields const fields = ["inlet", "oil", "water", "gas", "sepVol"]; fields.forEach(field => { const td
= document.createElement("td"); const input = document.createElement("input"); input.type = "number";
input.className = "editable-cell"; input.value = train[field]; input.step = field === "gas" ? "0.1" : field
=== "sepVol" ? "1" : "0.01"; input.min = "0"; input.id = `input-${idx}-${field}`;
input.addEventListener("input", function () { const val = parseFloat(this.value) || 0; trainData[idx][field]
= val; renderResultsTable(); renderKPIs(); }); input.addEventListener("focus", function () { this.select();
}); td.appendChild(input); tr.appendChild(td); }); tbody.appendChild(tr); }); } // ===== RENDER RESULTS
TABLE ===== function renderResultsTable() { const tbody = document.getElementById("resultsTableBody");
tbody.innerHTML = "";
```

```
trainData.forEach(train => { const r = calcResults(train); const tr = document.createElement("tr");
tr.innerHTML = `<td style="color:#00f0ff;font-weight:700;">${train.id}</td> <td>${fmt(r.waterCut)}</td>
<td>${fmt(r.gor)}</td> <td>${fmt(r.inletM3)}</td> <td>${fmt(r.retTime)}</td> `; tbody.appendChild(tr); }); }
// ===== RENDER KPI CARDS ===== function renderKPIs() { const g = calcGlobals(); animateValue("kpiInletVal",
g.totInlet); animateValue("kpiOilVal", g.totOil); animateValue("kpiDemulVal", g.demulRate);
animateValue("kpiEnergyVal", g.gasEnergy); } // ===== ANIMATE KPI VALUES ===== const currentValues = {};
function animateValue(elementId, target) { const el = document.getElementById(elementId); if (!el) return;
const start = currentValues[elementId] || 0; currentValues[elementId] = target; const duration = 600; const
startTime = performance.now(); function step(now) { const elapsed = now - startTime; const progress =
Math.min(elapsed / duration, 1); // Ease out cubic const ease = 1 - Math.pow(1 - progress, 3); const current
= start + (target - start) * ease; el.textContent = fmt(current); if (progress < 1)
requestAnimationFrame(step); } requestAnimationFrame(step); } // ===== INITIALIZE =====
document.addEventListener("DOMContentLoaded", function () { renderInputTable(); renderResultsTable();
renderKPIs(); }); });
```

■ File: app.py

```
import streamlit as st
import plotly.graph_objects as go
st.set_page_config(page_title="Hammar IPF Dashboard", layout="wide", page_icon="■")
st.markdown(""" <style> body, .stApp { background-color: #0a0e1a; color: #e0e6f0; font-family: 'Segoe UI', sans-serif; } .block-container { padding: 1rem 2rem; } h1,h2,h3 { color: #00d4ff; } div[data-testid="stMetric"] { background: linear-gradient(135deg,#0d1b2e,#1a2f4e); border: 1px solid #00d4ff44; border-radius: 12px; padding: 15px; } div[data-testid="stMetricValue"] { color: #00ff88 !important; font-size: 1.6rem !important; } div[data-testid="stMetricLabel"] { color: #8ab4d4 !important; } .stTabs [data-baseweb="tab"] { color: #8ab4d4; background: transparent; } .stTabs [aria-selected="true"] { color: #00d4ff !important; border-bottom: 2px solid #00d4ff; } hr { border-color: #1e3a5f; } .st-emotion-cache-1v0mbdj, .st-emotion-cache-z5fcl4 { background: #0d1424; } </style> """, unsafe_allow_html=True)
# ===== HEADER =====
st.markdown(""" <div style='background:linear-gradient(90deg,#050d1a,#0a1f3d,#050d1a); border-bottom:2px solid #00d4ff44; padding:25px 20px; margin-bottom:20px; border-radius:10px;'> <h1 style='margin:0; font-size:2.4rem; color:#00d4ff; letter-spacing:2px;'> ■■ HAMMAR IPF DASHBOARD </h1> <p style='color:#8ab4d4; margin:8px 0 0 0; font-size:1rem;'> Gas Oil Separation Plant – Integrated Production Facility | SCADA v2.4<br> <span style='color:#00ff88; font-size:0.85rem;'> <span style='color:#00ff88; font-size:0.85rem;'> ● LIVE | API-12J / API-14.1 / ASTM-D4007 / GPSA Compliant </span> </p> </div> """, unsafe_allow_html=True)
# ===== VIDEO =====
st.markdown("""### ■ Process Simulation – 3-Phase Separator (API-12J)""")
st.video("https://www.youtube.com/watch?v=JmKxN0s3-qQ")
st.markdown("---")
# ===== DEFAULT DATA =====
TRAINS = ["TRAIN 1", "TRAIN 2", "TRAIN 3", "TRAIN 4"]
DEFAULTS = { "Status": ["ON", "ON", "ON", "STOP"], "Inlet Fluid (bpd)": [31731.52, 46508.43, 51067.11, 0.0], "Oil Prod (bpd)": [22513.10, 37796.91, 39528.68, 0.0], "Water Prod (bpd)": [9218.42, 8711.52, 11538.43, 0.0], "Total Gas (MMSCFD)": [64.8, 36.1, 35.9, 0.0], "Separator Vol (m3)": [60.0, 60.0, 60.0, 60.0] }
# ===== LIVE DATA EDITOR =====
st.markdown("""### ■■ Live Production Data – Editable""")
st.info("■ Edit any value below. All charts and calculations update in real time.")
cols = st.columns([1.2, 1, 1.3, 1.3, 1.3, 1.3, 1.3])
```

```
headers = ["Train", "Status", "Inlet (bpd)", "Oil (bpd)", "Water (bpd)", "Gas (MMSCFD)", "Sep Vol (m3)"]
for col, h in zip(cols, headers): col.markdown(f"<div style='color:#00d4ff; font-size:0.8rem; font-weight:700; padding:4px 0;'>{h}</div>", unsafe_allow_html=True)
data = {}
for key in DEFAULTS: data[key] = list(DEFAULTS[key])
inlet, oil, water, gas, sep_vol, statuses = [], [], [], [], [], []
for i, train in enumerate(TRAINS): row_cols = st.columns([1.2, 1, 1.3, 1.3, 1.3, 1.3, 1.3])
row_cols[0].markdown(f"<div style='color:#00ff88; font-weight:bold; padding-top:8px;'>{train}</div>", unsafe_allow_html=True)
s = row_cols[1].selectbox("", ["ON", "STOP"], index=0 if DEFAULTS["Status"][i]=="ON" else 1, key=f"st_{i}", label_visibility="collapsed")
inf = row_cols[2].number_input("", value=DEFAULTS["Inlet Fluid (bpd)"][i], min_value=0.0, key=f"inf_{i}", label_visibility="collapsed", format="%.2f")
op = row_cols[3].number_input("", value=DEFAULTS["Oil Prod (bpd)"][i], min_value=0.0, key=f"op_{i}", label_visibility="collapsed", format="%.2f")
wp = row_cols[4].number_input("", value=DEFAULTS["Water Prod (bpd)"][i], min_value=0.0, key=f"wp_{i}", label_visibility="collapsed", format="%.2f")
tg = row_cols[5].number_input("", value=DEFAULTS["Total Gas (MMSCFD)"][i], min_value=0.0, key=f"tg_{i}", label_visibility="collapsed", format="%.2f")
sv = row_cols[6].number_input("", value=DEFAULTS["Separator Vol (m3)"][i], min_value=0.0, key=f"sv_{i}", label_visibility="collapsed", format="%.1f")
statuses.append(s); inlet.append(inf); oil.append(op); water.append(wp); gas.append(tg); sep_vol.append(sv)
st.markdown("---")
# ===== CALCULATIONS =====
EPS = 1e-9
def safe_div(a, b): return a / (b + EPS)
wc = [safe_div(water[i], inlet[i]) * 100 for i in range(4)]
gor = [safe_div(gas[i] * 1.000_000, oil[i]) for i in range(4)]
m3d = [inlet[i] * 0.158987 for i in range(4)]
rt = [safe_div(sep_vol[i] * 1440, m3d[i]) for i in range(4)]
eff = [safe_div(oil[i] + water[i], inlet[i]) * 100 for i in range(4)]
load = [safe_div(inlet[i], sep_vol[i]) for i in range(4)]
# Calculated table
st.markdown("""### ■ Calculated Engineering Results (API-12J / API-14.1 / ASTM-D4007)""")
hdr = st.columns([1.2, 1.2, 1.4, 1.2, 1.6, 1.4, 1.4])
for col, h in zip(hdr, ["Train", "Water Cut (%)", "GOR (SCF/bbl)", "Inlet m³/d", "Retention (min)", "Efficiency (%)", "Load (bpd/m³)"]): col.markdown(f"<div style='color:#00d4ff; font-size:0.78rem; font-weight:700;'>{h}</div>", unsafe_allow_html=True)
rt_ok = lambda v: "■" if 3 <= v <= 5 else "■" if v > 0 else "■"
wc_ok = lambda v: "■" if v < 30 else "■" if v < 50 else "■"
eff_ok = lambda v: "■" if v >= 95 else "■" if v >= 80 else "■"
for i in range(4): row = st.columns([1.2, 1.2, 1.4, 1.2, 1.6, 1.4, 1.4])
clr = "#00ff88" if statuses[i]=="ON" else "#ff4444"
```

```
row[0].markdown(f"<span style='color:{clr}; font-weight:bold;'>{TRAINS[i]}</span>", unsafe_allow_html=True)
row[1].markdown(f"{wc_ok(wc[i])} **{wc[i]:.2f}%**")
row[2].markdown(f"{gor[i]:.0f}%**")
row[3].markdown(f"{m3d[i]:.0f}%**")
row[4].markdown(f"{rt_ok(rt[i])} **{rt[i]:.2f}%**")
row[5].markdown(f"{eff_ok(eff[i])} **{eff[i]:.1f}%**")
row[6].markdown(f"{load[i]:.0f}%**")
st.caption("■ Normal ■ Warning ■ Critical | Retention target 3-5 min (API-12J) | GOR per API-14.1")
st.markdown("---")
# ===== CHARTS =====
DARK_BG = "#0a0e1a"
PLOT_BG = "#0d1424"
FONT_CLR = "#8ab4d4"
BASE_LAYOUT = dict(template="plotly_dark", paper_bgcolor=DARK_BG, plot_bgcolor=PLOT_BG, font_color=FONT_CLR, margin=dict(t=50,b=30,l=30,r=30))
tab1, tab2, tab3, tab4 = st.tabs(["■■■ Production", "■ Water & GOR", "■ Separator", "■ Efficiency"])
with tab1: fig = go.Figure()
fig.add_trace(go.Bar(name="Oil (bpd)", x=TRAINS, y=oil, marker_color="#00ff88"))
fig.add_trace(go.Bar(name="Water (bpd)", x=TRAINS, y=water, marker_color="#0099ff"))
fig.add_trace(go.Bar(name="Gas x1000", x=TRAINS, y=[g*1000 for g in gas], marker_color="#ffaa00"))
fig.update_layout(**BASE_LAYOUT, barmode="group", title="Production per Train")
st.plotly_chart(fig, use_container_width=True)
fig2 = go.Figure()
fig2.add_trace(go.Bar(name="Oil",
```

```

x=TRAINS, y=oil, marker_color="#00ff88")) fig2.add_trace(go.Bar(name="Water", x=TRAINS, y=water,
marker_color="#0099ff")) fig2.update_layout(**BASE_LAYOUT, barmode="stack", title="Liquid Composition
(Stacked)") st.plotly_chart(fig2, use_container_width=True) with tab2: col1, col2 = st.columns(2) with col1:
wc_colors = ["#00ff88" if v < 30 else "#ffaa00" if v < 50 else "#ff4444" for v in wc] fig =
go.Figure(go.Bar(x=TRAINS, y=wc, marker_color=wc_colors, text=[f"{v:.1f}%" for v in wc],
textposition="outside")) fig.add_hline(y=30, line_dash="dash", line_color="#ffaa00",
annotation_text="Warning 30%", annotation_position="right") fig.add_hline(y=50, line_dash="dash",
line_color="#ff4444", annotation_text="Critical 50%", annotation_position="right")
fig.update_layout(**BASE_LAYOUT, title="Water Cut % (ASTM-D4007)") st.plotly_chart(fig,
use_container_width=True) with col2: fig = go.Figure(go.Bar(x=TRAINS, y=gor, marker_color="#ffaa00",
text=[f"{v:,.0f}" for v in gor], textposition="outside")) fig.update_layout(**BASE_LAYOUT, title="GOR
SCF/bbl (API-14.1)") st.plotly_chart(fig, use_container_width=True) total_oil_sum = sum(oil)

```

```

total_water_sum = sum(water) fig_pie = go.Figure(go.Pie( labels=["Total Oil", "Total Water"],
values=[total_oil_sum, total_water_sum], marker_colors=["#00ff88", "#0099ff"], hole=0.4,
textinfo="percent+label")) fig_pie.update_layout(**BASE_LAYOUT, title="Plant Liquid Split")
st.plotly_chart(fig_pie, use_container_width=True) with tab3: col1, col2 = st.columns(2) with col1:
rt_colors = ["#00ff88" if 3 <= v <= 5 else "#ffaa00" if v < 3 else "#ff4444" for v in rt] fig =
go.Figure(go.Bar(x=TRAINS, y=rt, marker_color=rt_colors, text=[f"{v:.1f} min" for v in rt],
textposition="outside")) fig.add_hrect(y0=3, y1=5, fillcolor="#00d4ff", opacity=0.08,
annotation_text="API-12J: 3-5 min", annotation_position="top left") fig.update_layout(**BASE_LAYOUT,
title="Retention Time – API-12J Compliance") st.plotly_chart(fig, use_container_width=True) with col2: fig =
go.Figure(go.Bar(x=TRAINS, y=m3d, marker_color="#aa44ff", text=[f"{v:.0f}" for v in m3d],
textposition="outside")) fig.update_layout(**BASE_LAYOUT, title="Inlet Volume Flow (m³/d)")
st.plotly_chart(fig, use_container_width=True) fig = go.Figure(go.Bar(x=TRAINS, y=load,
marker_color="#ff6699", text=[f"{v:.0f}" for v in load], textposition="outside"))
fig.update_layout(**BASE_LAYOUT, title="Separator Liquid Loading (bpd/m³)") st.plotly_chart(fig,
use_container_width=True) with tab4: eff_colors = ["#00ff88" if v >= 95 else "#ffaa00" if v >= 80 else
"#ff4444" for v in eff] fig = go.Figure(go.Bar(x=TRAINS, y=eff, marker_color=eff_colors, text=[f"{v:.1f}%"
for v in eff], textposition="outside")) fig.add_hline(y=95, line_dash="dash", line_color="#00ff88",
annotation_text="Target >= 95%", annotation_position="right") fig.update_layout(**BASE_LAYOUT,
title="Separator Efficiency (%)") st.plotly_chart(fig, use_container_width=True) cats = ["Low Water Cut",
"Low GOR", "Retention OK", "Efficiency", "Throughput"] fig_radar = go.Figure() for i, train in
enumerate(TRAINS): if statuses[i] == "ON" and inlet[i] > 0: vals = [ max(0, 100 - wc[i]), max(0, 100 -
min(gor[i]/30, 100)), max(0, min(rt[i]/5*100, 100)), eff[i], min(inlet[i]/60000*100, 100), ]
fig_radar.add_trace(go.Scatterpolar( r=vals + [vals[0]], theta=cats + [cats[0]], name=train, fill="toself",
opacity=0.65)) fig_radar.update_layout(**BASE_LAYOUT, title="Train Performance Radar",
polar=dict(bgcolor=PLOT_BG)) st.plotly_chart(fig_radar, use_container_width=True) st.markdown("---") #
===== SURFACE CONTROL & KPI =====

```

```

st.markdown("### ■ Surface Control & Chemical Injection System") total_inlet = sum(inlet) total_oil =
sum(oil) total_water = sum(water) total_gas = sum(gas) demulsifier = ((total_inlet * 158.98) * (15 /
1_000_000)) / 24 scale_inh = ((total_inlet * 158.98) * (10 / 1_000_000)) / 24 corr_inh = ((total_inlet *
158.98) * (8 / 1_000_000)) / 24 gas_energy = total_gas * 1000 gas_gj = total_gas * 1055.056 st.markdown("####
■ Plant KPI Totals") c1, c2, c3, c4 = st.columns(4) c1.metric("Total Inlet Fluid", f"{total_inlet:,.0f}
bpd", f"{total_inlet*0.158987:,.0f} m³/d") c2.metric("Total Oil Produced", f"{total_oil:,.0f} bpd",
f"{safe_div(total_oil, total_inlet)*100:.1f}% of inlet") c3.metric("Demulsifier Rate", f"{demulsifier:.2f}
L/hr", "@15 ppm – API/NACE") c4.metric("Gas Thermal Energy", f"{gas_energy:,.0f} MMBTU/D", f"{gas_gj:,.0f}
MJ/D") st.markdown("#### ■ Chemical Injection Details") c1, c2, c3, c4 = st.columns(4) c1.metric("Scale
Inhibitor", f"{scale_inh:.2f} L/hr", "@10 ppm") c2.metric("Corrosion Inhibitor", f"{corr_inh:.2f} L/hr", "@8
ppm") c3.metric("Total Water Prod", f"{total_water:,.0f} bpd", f"{total_water*0.158987:,.0f} m³/d")
c4.metric("Total Gas Prod", f"{total_gas:.2f} MMSCFD", f"{total_gas*28316.8:,.0f} m³/d") chems =
["Demulsifier", "Scale Inhibitor", "Corrosion Inhibitor"] rates = [demulsifier, scale_inh, corr_inh]
fig_chem = go.Figure(go.Bar(x=chems, y=rates, marker_color=["#00ff88", "#0099ff", "#ffaa00"],
text=[f"{r:.3f} L/hr" for r in rates], textposition="outside")) fig_chem.update_layout(**BASE_LAYOUT,
title="Chemical Injection Rates (L/hr)") st.plotly_chart(fig_chem, use_container_width=True)
st.markdown("---") st.success("■ All systems are stable. Mass balance is automatically updated based on your
inputs.") st.caption("Standards: API-12J | API-14.1 | API-14.5 | ASTM-D4007 | API-RP-19B | GPSA Engineering
Data Book")

```

■ File: sw.js

```
var CACHE='hammar-scada-v3'; var FILES=['index.html','styles.css','login.css','core.js','charts.js','feature  
s.js','login.js','manifest.json','sw.js','icon-192.png','icon-512.png','■■■■ ■■■■■■■■.jpg']; self.addEvent  
Listener('install',function(e){self.skipWaiting();e.waitUntil(caches.open(CACHE).then(function(c){return  
c.addAll(FILES)}))});  
self.addEventListener('fetch',function(e){e.respondWith(fetch(e.request).catch(function(){return  
caches.match(e.request)}))});  
self.addEventListener('activate',function(e){e.waitUntil(caches.keys().then(function(ks){return  
Promise.all(ks.filter(function(k){return k!==CACHE}).map(function(k){return caches.delete(k)}))})));});
```

■ File: manifest.json

```
{ "name": "Hammar IPF - Smart SCADA", "short_name": "Hammar SCADA", "description": "Advanced GOSP Simulation & Engineering Calculator", "start_url": "./index.html", "display": "standalone", "orientation": "any", "background_color": "#05080f", "theme_color": "#00f0ff", "icons": [ {"src": "icon-192.png", "sizes": "192x192", "type": "image/png"}, {"src": "icon-512.png", "sizes": "512x512", "type": "image/png"} ] }
```

■ File: start_online.vbs

```
Set WshShell = CreateObject("WScript.Shell") Dim fso Set fso = CreateObject("Scripting.FileSystemObject") '
Delete old log if exists If fso.FileExists("cloudflared.log") Then fso.DeleteFile "cloudflared.log", True
End If ' Start Python server hidden WshShell.Run "cmd /c start /b python -m http.server 8080", 0, False '
Start Cloudflared hidden and redirect to log WshShell.Run "cmd /c .\cloudflared.exe tunnel --url
http://localhost:8080 2> cloudflared.log", 0, False ' Wait for the log file to be created and contain the URL
WScript.Sleep 3000 url = "" For i = 1 To 15 If fso.FileExists("cloudflared.log") Then On Error Resume Next
Set file = fso.OpenTextFile("cloudflared.log", 1) If Err.Number = 0 Then content = file.ReadAll() file.Close
' Find trycloudflare.com startPos = InStr(content, "https://") If startPos > 0 Then endPos = InStr(startPos,
content, ".trycloudflare.com") If endPos > 0 Then url = Mid(content, startPos, endPos - startPos + 18) Exit
For End If End If End If On Error GoTo 0 End If WScript.Sleep 2000 Next If url <> "" Then ' Open in default
browser WshShell.Run url ' Copy to clipboard (requires IE or HTMLFile trick, but simple MsgBox is safer)
MsgBox "██ ██████████ ████████ ████████! ██████████ █████ ██████████ ██████████." & vbCrLf & vbCrLf & "██████
██████████ ██████████ ██████████ ██████████:" & vbCrLf & url & vbCrLf & vbCrLf & "██ ██████████ ██████████ ██████████. ██████████
██████████ ██████████ ██████████ ██████████.", 64, "Hammar IPF Online" Else MsgBox "██████ ██████████ ██████████ ██████████
██████████. ██████████ ██████████ ██████████ ██████████ ██████████.", 16, "██████" End If
```

■ File: Final_Push.bat

```
@echo off title Hammar IPF - Professional Deployment echo
===== echo HAMMAR IPF - AUTOMATED GITHUB DEPLOYMENT echo
===== echo. :: Initialize git if not already if not exist
.git ( echo [1/4] Initializing Git... git init ) else ( echo [1/4] Git already initialized. ) :: Configure
remote (Using your GitHub URL) echo [2/4] Configuring Remote Origin... git remote remove origin >nul 2>&1 git
remote add origin https://github.com/sajadkhano/HammarIPF.git :: Stage and Commit echo [3/4] Staging files
and committing changes... git add . git commit -m "Professional SCADA Dashboard Release v1.0 - Full Features
& Engineering Calc" :: Push to GitHub echo [4/4] Pushing to GitHub (Main Branch)... echo. echo NOTE: A
browser window or login prompt may appear for GitHub authentication. echo. git branch -M main git push -u
origin main echo. echo ===== echo SUCCESS: Project uploaded
to GitHub! echo Your final professional link will be: echo https://sajadkhano.github.io/HammarIPF/ echo.
echo Don't forget to enable GitHub Pages in settings! echo
===== pause
```

■ File: .gitignore

```
cloudflared.exe cloudflared.log *.zip *.log __pycache__/ .vscode/
```